

CEHY22301 | Dynamic Simulation for Process Design and Performance Optimization using Aspen Hysys® Solution



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Convert a Steady-state Model into a Dynamic Model

Workshop

Files can be found in folder: C:_Workshops\EHY\EHY223\EHY223.12.01 on your cloudshare environment

Objective

In this workshop, you will convert a simple steady-state model into a dynamic one by following the standard procedure. You will enter equipment geometry data and make dynamic specifications before jumping straight to dynamic mode.

In this workshop, we will focus on the fundamental concepts underlying the use and configuration of Aspen HYSYS Dynamics.

Description

This module explains the process of converting a steady-state simulation into a dynamic one. The process is not difficult, but key procedures should be followed to ensure a smooth transition from steady state to dynamic mode.

For process design activities, users typically enter basic design information and Aspen HYSYS estimates reasonable defaults for the detailed equipment information. Typically, these basic design parameters can be found on the Design tab of unit operations. Any detailed equipment information estimated by Aspen HYSYS Dynamics is shown in blue italicized text. Generally, this detailed information can be found on either the Rating or Dynamics tabs of the unit operation property view.

This workshop includes the following tasks:

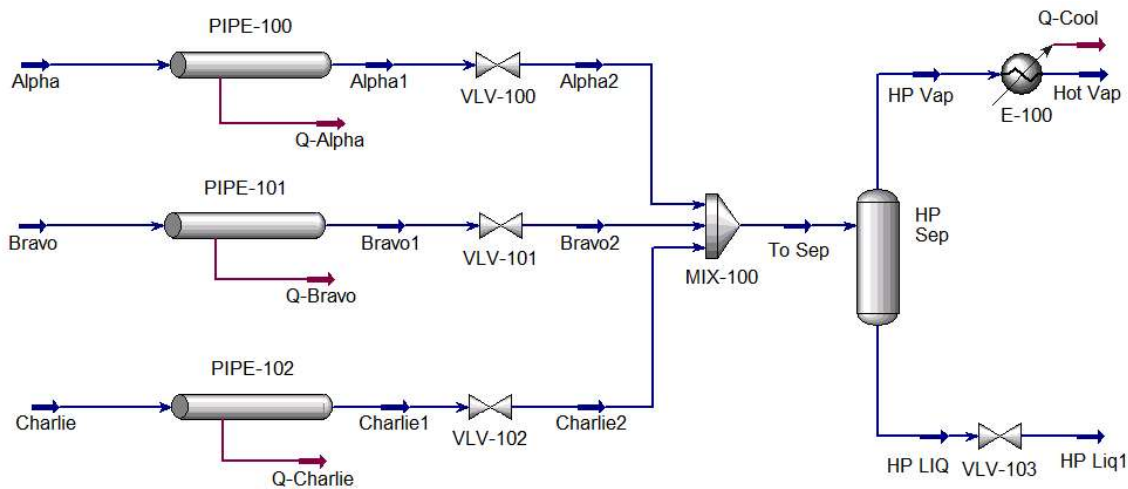
- Task 1 – Equipment Sizing.
- Task 2 – Make Pressure/Flow Specifications.
- Task 3 – Move to Dynamic Mode.

Task 1 – Equipment Sizing

To have a dynamic model perform properly, many additional specifications will need to be made in steady-state. In addition, some operations must be added to the model while others will need to be removed. In general, the following should be considered before moving from steady-state to dynamic mode:

- Consider the addition of unit operations that cause a pressure drop in the process.
- Enter geometry data for all process equipment.
- Adjust pressure specs for column models.
- Remove logical operations (e.g., Recycle blocks).
- Add controllers to ensure that a stable operation can be attained.
- Make either flow or pressure specs on all boundary (inlets / outlets) streams.

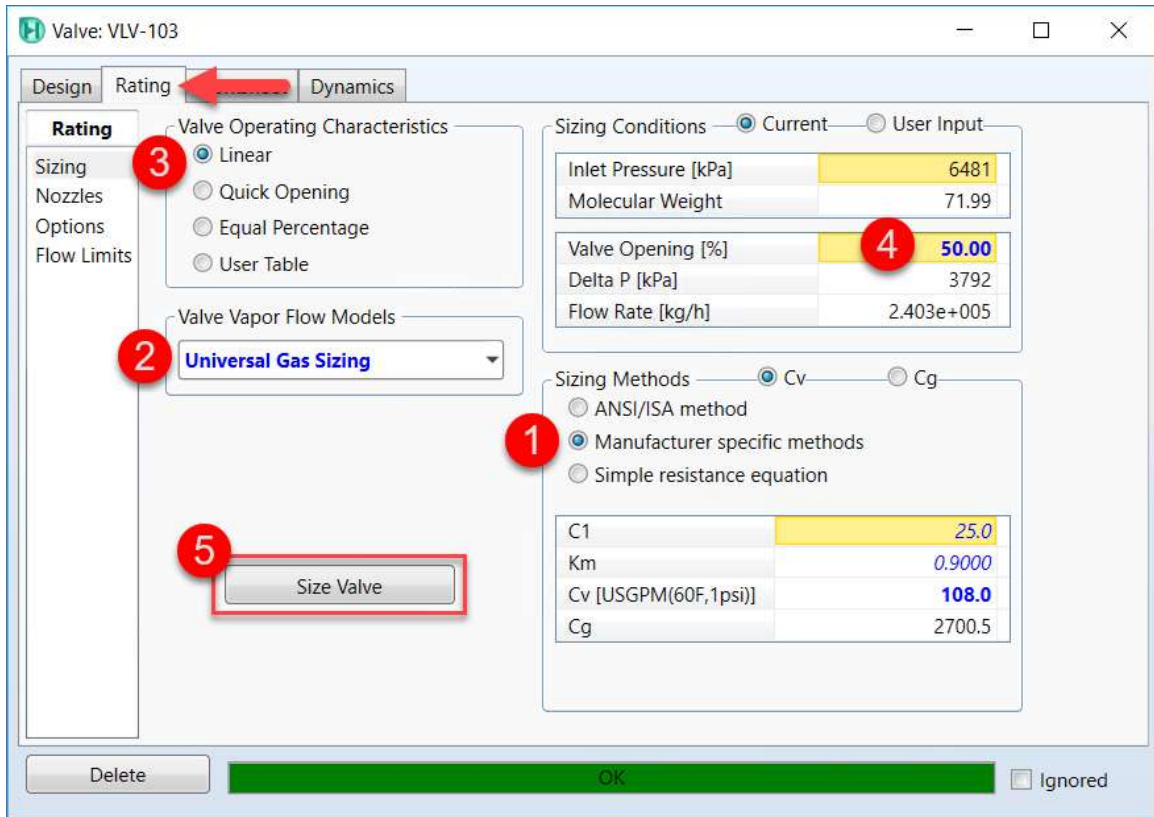
1. Open the file **01-SteadyStateStarter.hsc**.
2. Save the file as **01-PrepareForDynamic.hsc**.



You will begin by properly sizing the unit operations in the steady-state model. Begin with the four valve operations.

3. Double-click on **VLV-103**.
4. On the **Rating** tab of the **Valve** property view, under the **Sizing Methods** section, select the **Manufacturer specific methods** option and choose **Universal Gas Sizing** as the **Valve Vapor Flow Model**. On the **Valve Operating Characteristics** section, set the valve flow characteristics as **Linear**.

- Set the **Valve Opening (%)** in the **Sizing Conditions** section as **50%**. Click on the **Size Valve** button to complete the sizing calculation.



Valve: VLV-103

Design Rating Dynamics

Rating

Sizing

Nozzles

Options

Flow Limits

Valve Operating Characteristics

☒ Linear

☐ Quick Opening

☐ Equal Percentage

☐ User Table

Valve Vapor Flow Models

Universal Gas Sizing

Sizing Conditions ☒ Current ☐ User Input

Inlet Pressure [kPa] 6481

Molecular Weight 71.99

Valve Opening [%] **50.00**

Delta P [kPa] 3792

Flow Rate [kg/h] 2.403e+005

Sizing Methods ☒ Cv ☐ Cg

☐ ANSI/ISA method

☒ Manufacturer specific methods

☐ Simple resistance equation

C1 25.0

Km 0.9000

Cv [USGPM(60F,1psi)] **108.0**

Cg 2700.5

Size Valve

Delete OK Ignored

In this case, the sizing calculation computes a **C_v** equal to **108 USGPM**. The valve will handle **240,300 kg/hr (529,800 lb/hr)** when the valve is **50%** open with a pressure drop of **3792 kPa (550 psi)**.

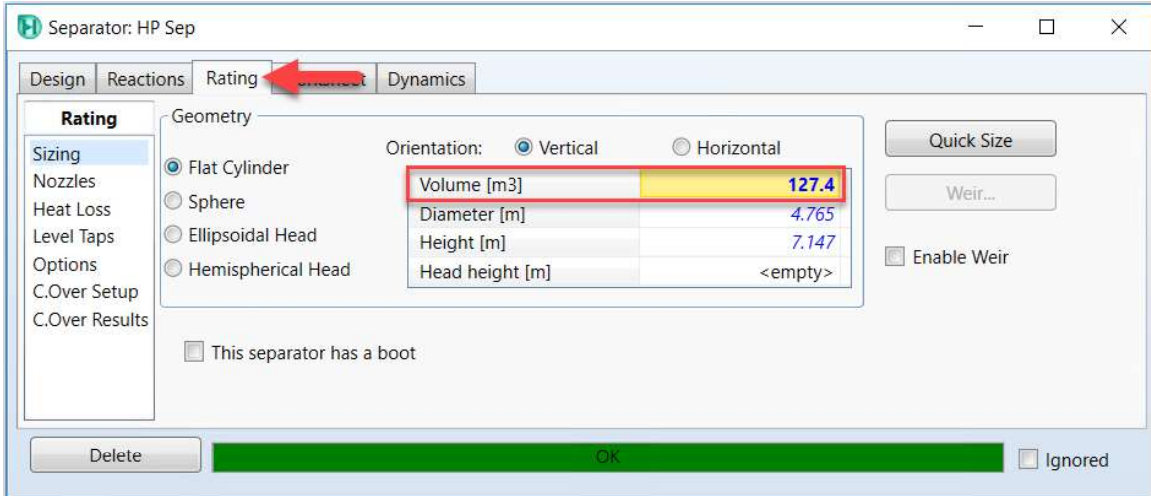
- Do the same for the other three valves (**VLV-100**, **VLV-101**, **VLV-102**).



QUESTION: What are the calculated **C_v** values? [Write your answer below]

Valve	Calculated C _v
VLV-100	
VLV-101	
VLV-102	

- Double-click on the **HP Sep** operation and go to the **Rating** tab.
- Enter a Volume of **127.4 m³ (4400 ft³)**.



Separator: HP Sep

Design Reactions **Rating** Dynamics

Rating

Geometry

Orientation: ☒ Vertical ☐ Horizontal

☒ Flat Cylinder

☐ Sphere

☐ Ellipsoidal Head

☐ Hemispherical Head

Volume [m3] 127.4

Diameter [m] 4.765

Height [m] 7.147

Head height [m] <empty>

☐ This separator has a boot

Quick Size

Weir...

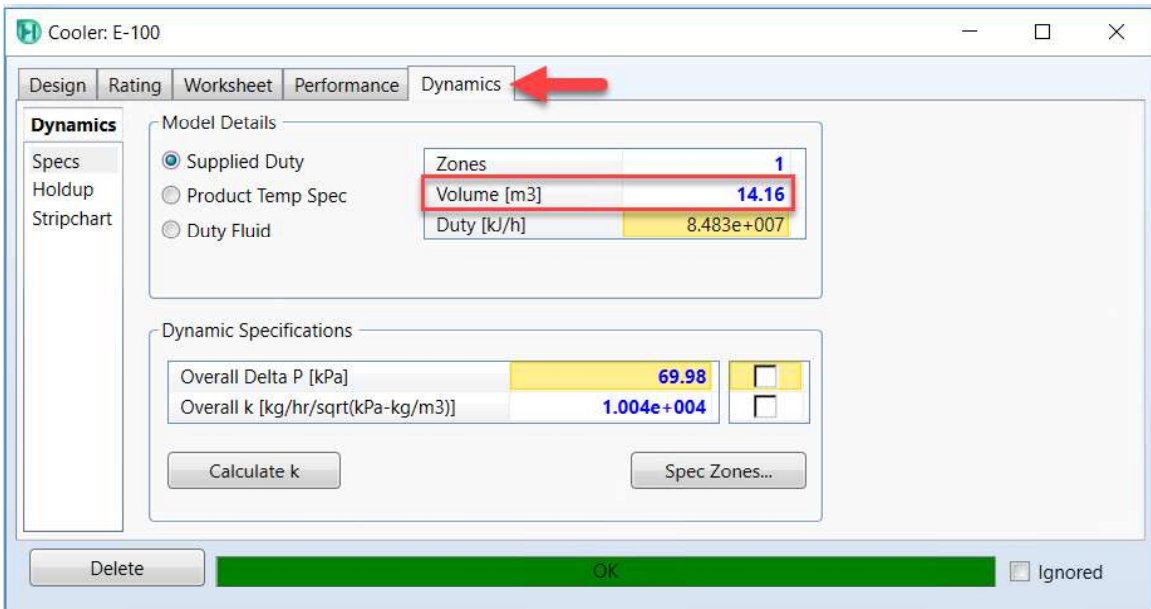
☐ Enable Weir

Delete OK Ignored

Unlike separators, several unit operations have default volume values (e.g., Heat Exchangers, Heaters, Coolers). When adding this type of unit operations to the flowsheet, make sure that reasonable equipment volumes are specified.

9. Next, you will size the **E-100** cooler. Double-click on the operation and go to the **Dynamics** tab.

10. Enter a **Volume** of **14.16 m3 (500 ft3)**.



Cooler: E-100

Design Rating Worksheet Performance **Dynamics**

Dynamics

Specs

Holdup

Stripchart

Model Details

☒ Supplied Duty

☐ Product Temp Spec

☐ Duty Fluid

Zones 1

Volume [m3] 14.16

Duty [kJ/h] 8.483e+007

Dynamic Specifications

Overall Delta P [kPa] 69.98

Overall k [kg/hr/sqrt(kPa-kg/m3)] 1.004e+004

Calculate k

Spec Zones...

Delete OK Ignored

The pipe segment operations geometry is defined with both length and diameter; these sizing inputs are enough. As for the mixer block, the Equalize All pressure specification should be made (which is consistent with real life); it is already specified this way in the starter file.

11. Now all operations have been properly sized. Save **your case**.

Task 2 – Make Pressure/Flow Specifications

Aspen HYSYS Dynamics considers the flowsheet as a boundary value problem, all internal streams will be solved simultaneously (using an equation-oriented approach). Also, Aspen HYSYS Dynamics will need the system to be a square one (degrees of freedom = 0), so enough (and consistent) pressure/flow specifications must be made to satisfy this condition. The possible pressure or flow specifications for a flowsheet include:

- Pressure specification on a boundary (inlet / outlet) stream.
- Flow specification on a boundary (inlet / outlet) stream.
- Fixed pressure drop across a unit operation.
- Resistance equations (valves).
- Conductance equations (process equipment).
- Pressure/flow equation.

Pressure/flow stream specifications are made on the Specs page of the Dynamics tab of any material stream. Typically, boundary streams have just a single dynamic spec, either pressure or flow, but not both. In this task, a single pressure spec will be used for all boundary streams.

Dynamic specifications for unit operations are also generally found on the Dynamics tab of the unit operation property view. The following table contains some guidelines that will help guarantee a consistent and properly specified flowsheet to be run in dynamic mode.

Dynamic Specifications	
Boundary Streams	Insert a valve on all boundary streams (feed/product streams) within the Flowsheet that are not connected to conductance devices (such as heat exchangers, coolers, heaters).
Pressure Specifications	Place a pressure specification on all boundary streams (feed/product streams) within the Flowsheet.
Distillation Column	Distillation columns with condensers require an extra specification around the condenser. Make a flow specification for the reflux flow.
Valves	Use the pressure/flow relationship as the dynamic specification for a valve.
K value	Use the overall K value as the dynamic specification for coolers, heaters, and heat exchangers and LNG exchangers.
Pressure Gradients	Be sure to account for pressure gradients throughout the Flowsheet and specify reasonable pressure drops/rises in the Flowsheet. Pressure differentials are the driving force for flow through the process Flowsheet.
Column Internals Analysis	Use the Column Internals Analysis to estimate the column geometry and pressure profile.
Mixers	Use the Equalize All option as the pressure specification for mixers.
Tees	Remove Use Splits as Dynamic Flow Specs on tees.
Rotating Equipment (Pumps, Compressors, Expanders)	Use Efficiency and either Head or Pressure Rise as dynamic specifications for rotating Equipment. Compressor and Pump Curves , if available, make excellent dynamic specifications.
Hold-ups	Be sure to properly size equipment with hold-ups.
Dynamic specifications can only be modified when the Integrator is stopped. Once the Integrator is started the value of the dynamic specification can be changed (its value displays in blue), but the choice of dynamic specification cannot be changed.	

Based on the previous table, we can make the following observations:

- The boundary streams are Alpha, Bravo, Charlie, HotVap, and HP Liq1.
- HP Liq1 is connected to valve VLV-103, so the addition of a valve is not required at all.
- HotVap is connected to a conductance equation, E-100, so the addition of a valve is not required either.
- The upstream side has three pipe segments and three valves to control flowrate, so there is no need for any additional resistance / conductance equation.

All boundary streams in the flowsheet must have either a dynamic pressure or flow specification. The boundary streams in this model are: Alpha, Bravo, Charlie, HotVap, and HPLiq1. Use the Dynamic P/F Specs color scheme to quickly identify dynamic specs in streams. This colour scheme can be customized using the drop-down menu in the Flowsheet/Modify contextual tab of the Aspen HYSYS ribbon. With the Dynamic P/F Specs colour scheme, you can use the following colour coding to ensure that you have properly specified your streams:

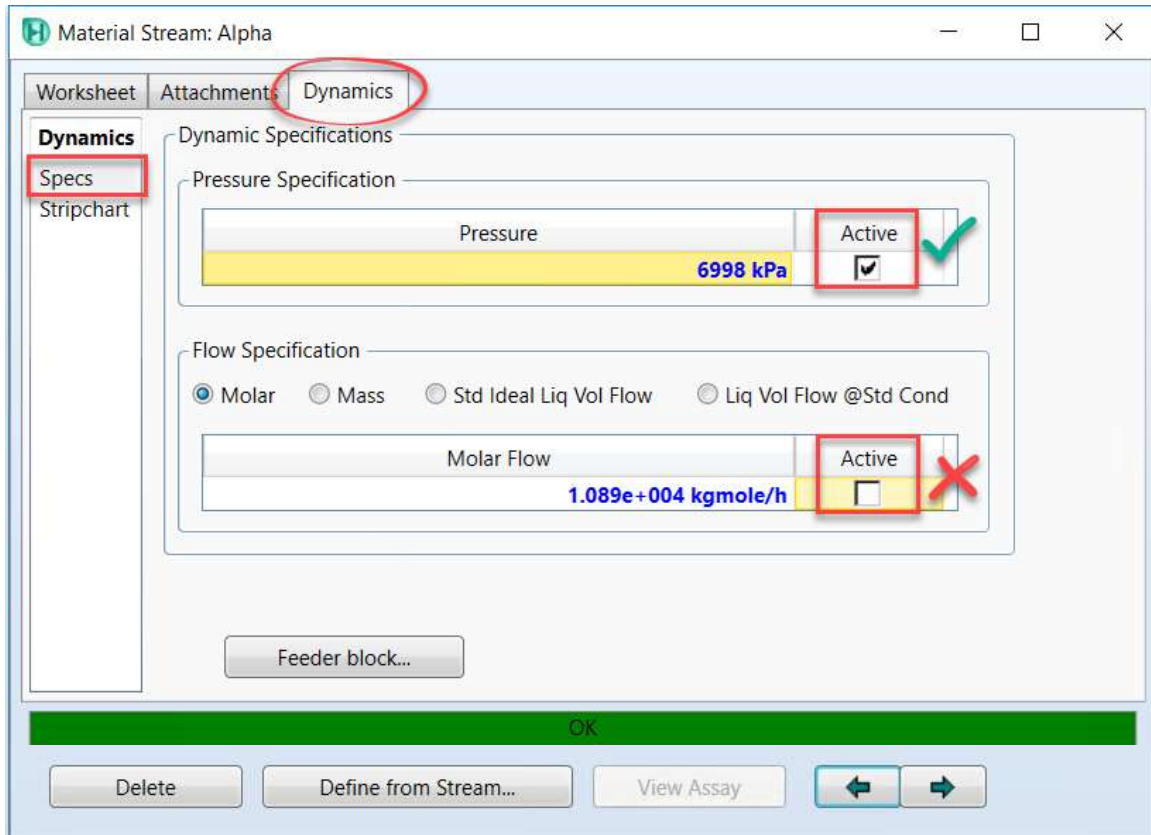
- Blue stream = No dynamic specifications
- Green stream = Dynamic pressure specification
- Yellow stream = Dynamic flow specification
- Red stream = Both dynamic pressure and flow specifications
- Grey stream = Energy stream

12. Go to **Flowsheet/Modify** ribbon. Change the color scheme to **Dynamic P/F Spec**.



After doing this you will notice that the color of the feed streams is red. That will tell you that these streams have both dynamic pressure and flow specs. As mentioned before, we need to define a single dynamic spec for boundary streams.

13. Double-click on the **Alpha** stream arrow and go to the **Dynamics** tab. Make sure that the **Pressure Specification** is **Active**. Uncheck the flow spec if checked.



14. **Activate** the dynamic **pressure** specification for the **Bravo, Charlie, HotVap,** and **HP Liq1** material streams as well. At the same time, make sure that the dynamic flow specification is NOT active in any of these streams.
15. Double-click on **Alpha2** and go to the Dynamics tab. Uncheck any checked specs. Remember that only the boundary streams should have a dynamic spec.



Note: As a shortcut, instead of going to each stream's **Dynamics** tab, open each **unit operation** and go to the **Worksheet | PF Specs** tab to view the pressure flow specifications for each of the connected streams.

Cooler: E-100

Design Rating **Worksheet** Performance Dynamics

Worksheet

Conditions
Properties
Composition
PF Specs

	HP Vap	Hot Vap
Pressure Spec Active	☐	☒
Pressure [kPa]	6481	6411
Flow Spec Active	☐	☐
Flow Spec Type	Molar Flow	Molar Flow
Molar Flow [kgmole/h]	2.933e+004	2.933e+004
Mass Flow [kg/h]	6.225e+005	6.225e+005
Std Ideal Liq Vol Flow [m3/h]	1734	1734

The '**K**' variable is the conductance-to-flow constant for unit operations such as Heaters, Coolers, Heat Exchangers, etc. Its value is calculated based on the current steady-state pressure drop, density, and flowrate going through the unit operation.

16. Double-click on the **E-100** Cooler. On the **Dynamics | Specs** page, click on the **Calculate k** button.

17. Once the **K** value has been computed, activate the **Overall k** checkbox and make sure that the **Overall Delta P** checkbox is clear.

Cooler: E-100

Design Rating Worksheet Performance **Dynamics**

Dynamics

Specs
Holdup
Stripchart

Model Details

☒ Supplied Duty
☐ Product Temp Spec
☐ Duty Fluid

Zones	1
Volume [m3]	14.16
Duty [kJ/h]	8.483e+007

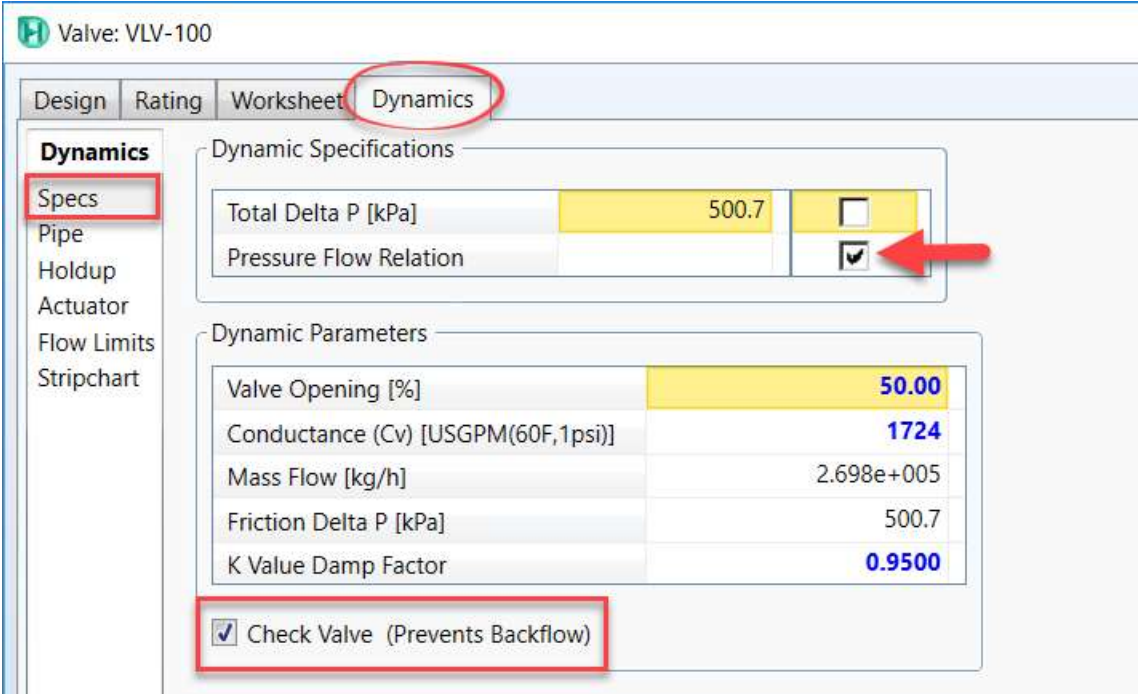
Dynamic Specifications

Overall Delta P [kPa]	69.98	☐
Overall k [kg/hr/sqrt(kPa-kg/m3)]	1.004e+004	☒

Calculate k Spec Zones...

It is useful to verify that all other resistance/conductance equations (in this case, the four valve operations) are using a pressure/flow relationship, rather than having a fixed dP. If pressure drop is fixed, then there won't be any relationship between pressure and flow at that point of the model, making the dynamic calculation inflexible (and that might lead to convergence failures). Take a moment to confirm that the four valve operations are set up properly before continuing.

18. Double-click on **VLV-100** and go to the **Dynamics | Specs** page. Make sure that the **Pressure Flow Relation** checkbox is checked and that the **Total Delta P** checkbox is **clear**. Activate the **Check Valve** option. This helps to prevent reverse flow in case of negative pressure gradient.



Valve: VLV-100

Design Rating Worksheet **Dynamics**

Dynamics

Specs

Pipe

Holdup

Actuator

Flow Limits

Stripchart

Dynamic Specifications

Total Delta P [kPa]	500.7	☐
Pressure Flow Relation		☒

Dynamic Parameters

Valve Opening [%]	50.00
Conductance (Cv) [USGPM(60F,1psi)]	1724
Mass Flow [kg/h]	2.698e+005
Friction Delta P [kPa]	500.7
K Value Damp Factor	0.9500

☒ Check Valve (Prevents Backflow)

19. **Repeat** this procedure for the other three valves.
20. Double-click on **PIPE-100**. Go to the **Dynamics** tab. Keep all the defaults. Note that the Simple Pipe Friction Model Method will be used by default. It means that the model selected under the **Design | Parameters** page will not be used. If you wanted to use one of the multiphase correlations on this page, then you would need to choose the second option (Pipe Model Correlations). You will keep the default choice to keep it simple.



Note: You can always get more information about each specification cell in Aspen HYSYS by either hitting the F1 key or clicking on the Help button. Also, the Aspen Knowledge tool can be used under the **Resources** tab.

Pipe Segment: PIPE-100

Design Rating Worksheet Performance Flow Assurance Dynamics

Dynamics

Parameters
Holdup
Stripchart

Parameters

Pipe Flow Model

☒ Simple Pipe Friction Model Method
☐ Pipe Model Correlations

Pipe Friction Model **turbulent**

Pipe Holdup Type

☐ one dp calc/pipe
☒ one dp calc/segment

Model Holdup Volume ☒

Base Elevation of Inlet Relative to Ground **0.0000**

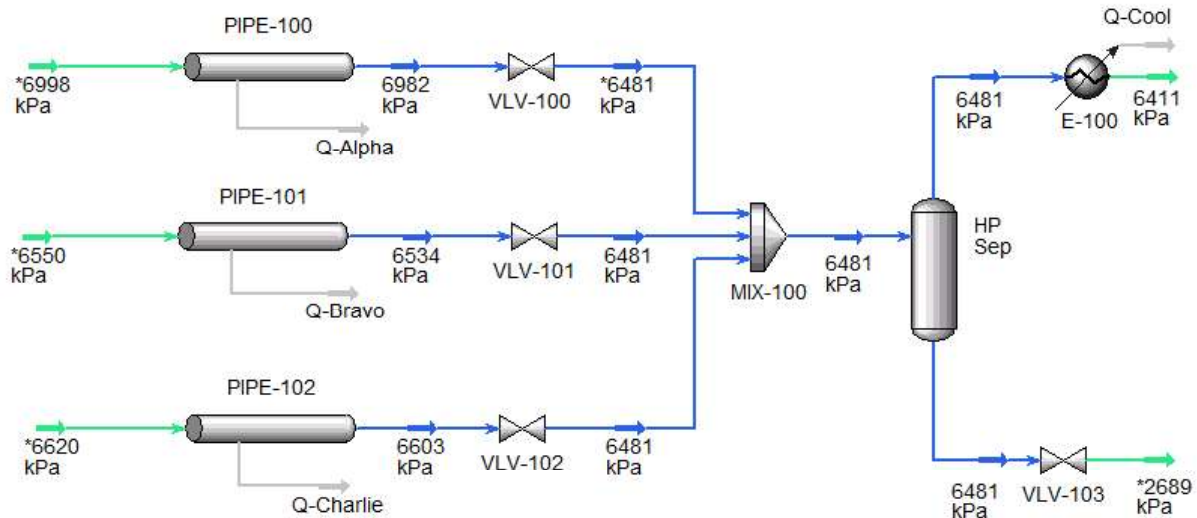
21. Save your file as: **01-ReadyForDynamic.hsc**

Task 3 – Move to Dynamic Mode

Once you have sized all unit operations, made all the proper dynamic specifications, and achieved a proper pressure profile across the process, the model should be ready for dynamic mode. Before moving to dynamic mode, confirm that the model is correctly set up.

22. Click on an empty place on the PFD. Press **Shift+P** to see the pressure values of all streams.

If any of the answers to the above questions are no, then fix the problem (ask the Instructor for help if needed).

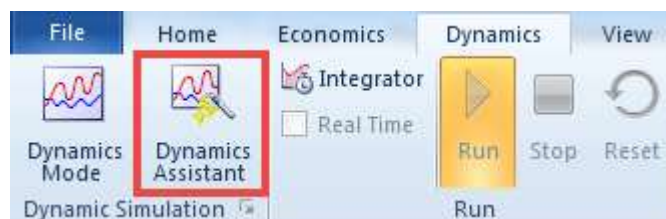


There is another way to confirm whether the steady-state model is ready to be brought to dynamic mode or not. This tool is the Dynamics Assistant.

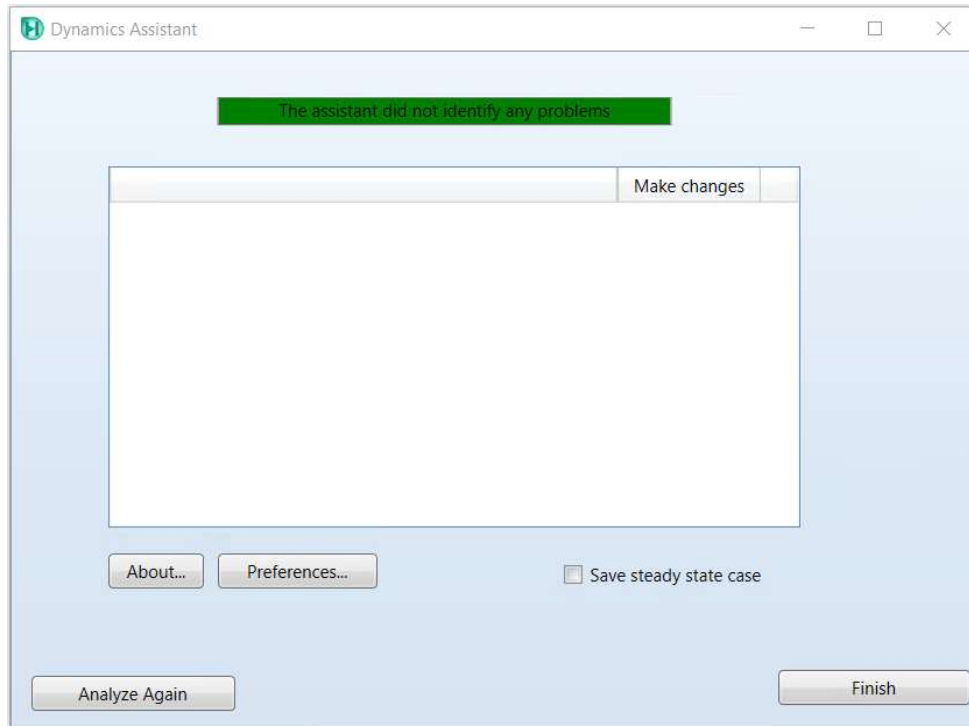
The Dynamics Assistant can suggest changes that will make sure that the model runs in dynamic mode. However, if your simulation deviates from the 'standard' setup, the Dynamics Assistant will list undesired changes that might not meet your expectations. Always look at any possibly listed suggestions and determine whether these are crucial changes or not. If not, you may ignore them.

The Dynamics Assistant will be activated when making the transition from steady state to dynamic mode and it will check the readiness of the model. You can also click on the Dynamics Assistant button to bring it up.

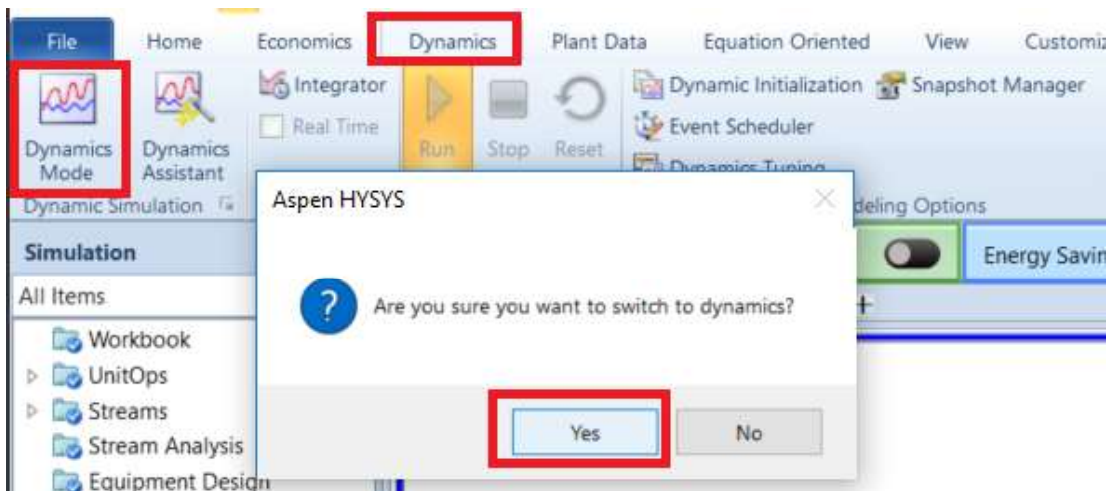
23. Go to the **Dynamics** tab on the ribbon and click on the **Dynamics Assistant** button.



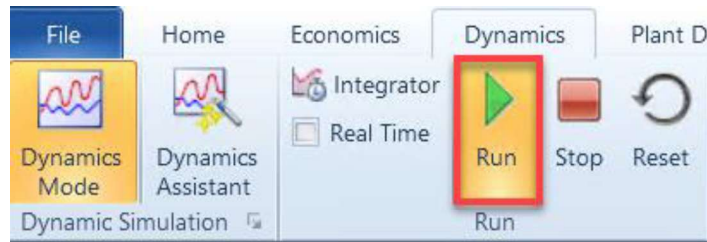
24. This time the Dynamics Assistant will not find any problems. The model is ready to be run in dynamic mode.



25. **Close** the Dynamics Assistant window.
26. The file is still in steady-state. You are about to move to dynamic mode. It is smart to always keep a copy of the steady-state model you are using to depart to dynamic mode.
27. **Save** the file.
28. Click on **Dynamics Mode** button and click on **Yes** when prompted to do so.



29. The file is in dynamic mode now, but it is not running. Save the file as **01_Dynamic.hsc** and then click on the **Run** button.

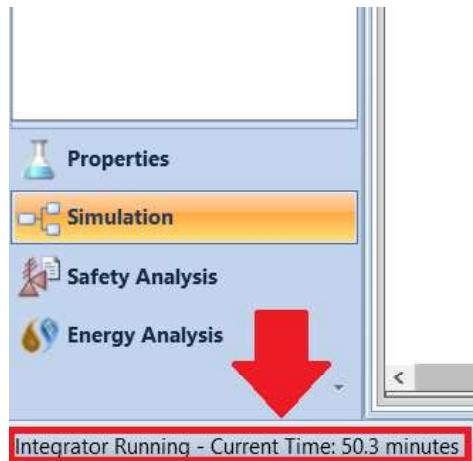


30. **Stop** the dynamic run after a while.

31. Save the file.



Note: A way to check if a simulation is running in Dynamic mode is to look at the lower left side of the screen, it should read either 'Integrator running' or 'Integrator Stopped'.



Add Controllers and Strip Charts

Workshop

Files can be found in folder:

C:_Workshops\EHY\EHY223\EHY223.12.01\02_ControllersStripCharts on your cloudshare environment

Objective

In this workshop, you will add PID controllers to the dynamic model and configure them to test process control strategies. To successfully implement PID controllers in Aspen HYSYS Dynamics, you will learn how to set them up and add strip charts to monitor the dynamic performance of key process variables.

Description

The dynamic model from the previous workshop has all the necessary specifications and inputs in place and can be successfully run-in dynamic mode now. However, without controllers, the model would run in an open-loop. The system would either stabilize or run out of control. Furthermore, there are no effective means of tracking process data as the dynamic model runs. To fix these deficiencies, this workshop will walk you through the procedure of adding and setting up controllers and strip charts.

This workshop includes the following tasks:

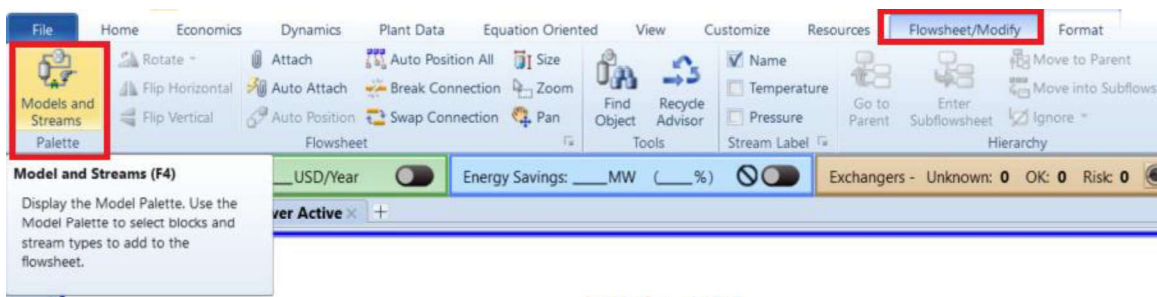
- Task 1 – Add PID Controllers.
- Task 2 – Build Strip Charts and Observe the Dynamic Behavior of the Model.
- Task 3 – Customize Strip Charts and Change the Controllers Set Point.
- Task 4 – Estimate PID Controller Tuning Parameters (Optional)

Task 1 – Add PID Controllers

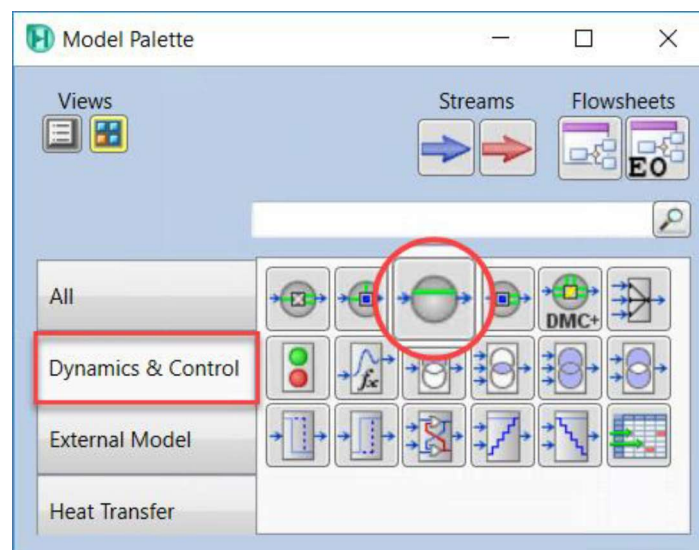
Controllers can be added to the flowsheet the exact same way as unit and logical operations are added. The Dynamics section of the Object Palette contains the available controller models. In this workshop you will use the PID Controller. Once a PID Controller has been added to the flowsheet, the following should be considered:

- Define the controller process variable source and output target object.
- Define both the minimum and maximum values for the process variable range.
- Select the correct controller action: reverse / direct.
- Enter the controller tuning parameters.
- Select the controller operating mode: Off, Indicator, Manual, or Automatic.

1. Open the case you saved at the end of the previous workshop: **01-Dynamic.hsc**.
2. If the model palette is not open, go to the **Flowsheet/Modify** tab and click on the **Models and Streams** button. The shortcut to launch the model palette is to hit the F4 key on your keyboard.

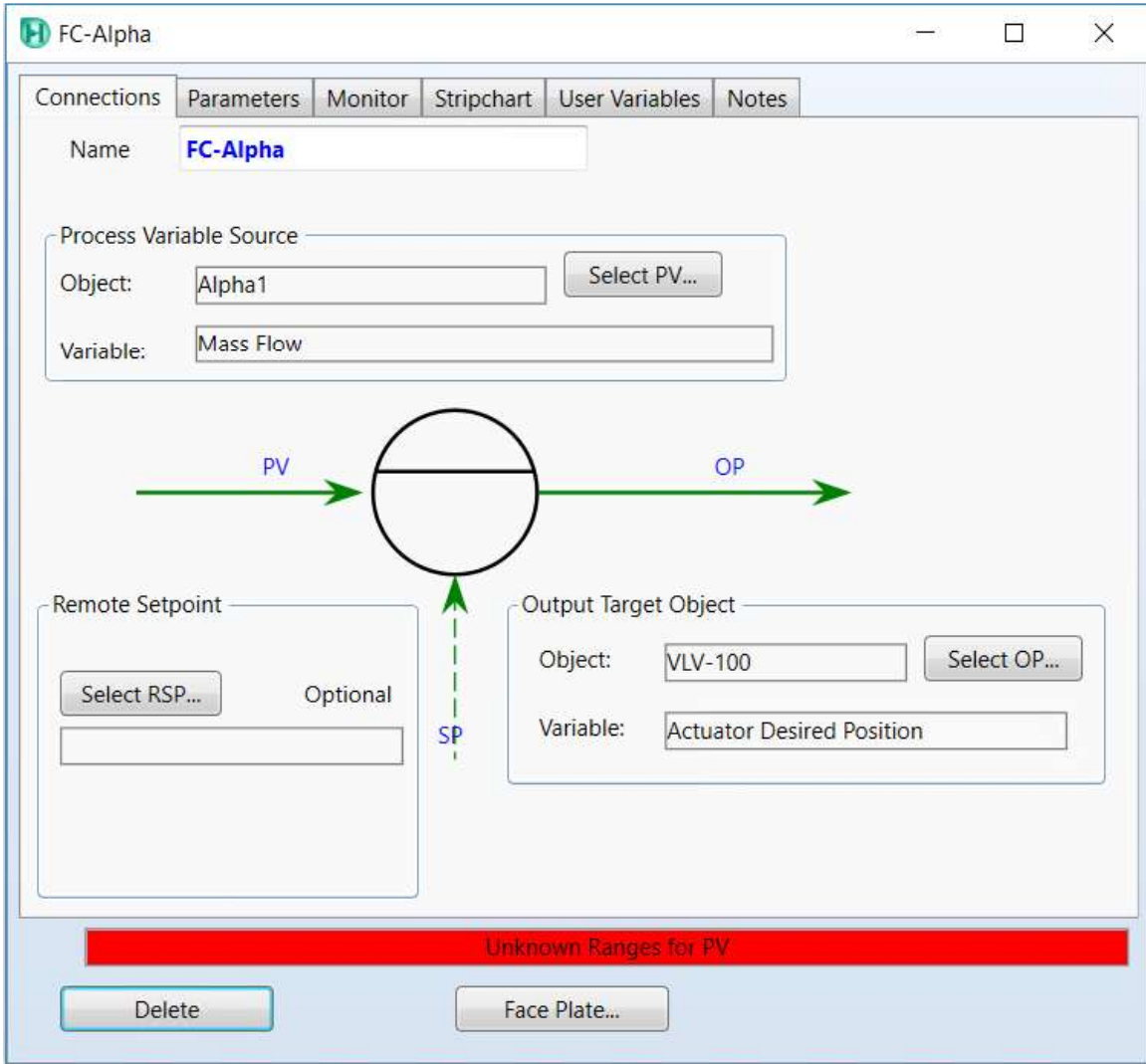


3. Add a **PID Controller** to the flowsheet from the **Dynamics & Control** section of the Model Palette.



4. Double click on the controller block and rename it as **FC-Alpha**.
5. Click on the **Select PV** button and select **Alpha1 | Mass Flow** as the **Process Variable Source** and finally click on **Select**.

6. Click on the OP button and select **VLV-100 | Actuator Desired Position** as the **Output Target Object** and click on **Select**.



FC-Alpha

Connections Parameters Monitor Stripchart User Variables Notes

Name: **FC-Alpha**

Process Variable Source

Object: Alpha1 Select PV...

Variable: Mass Flow

Remote Setpoint

Select RSP... Optional

Output Target Object

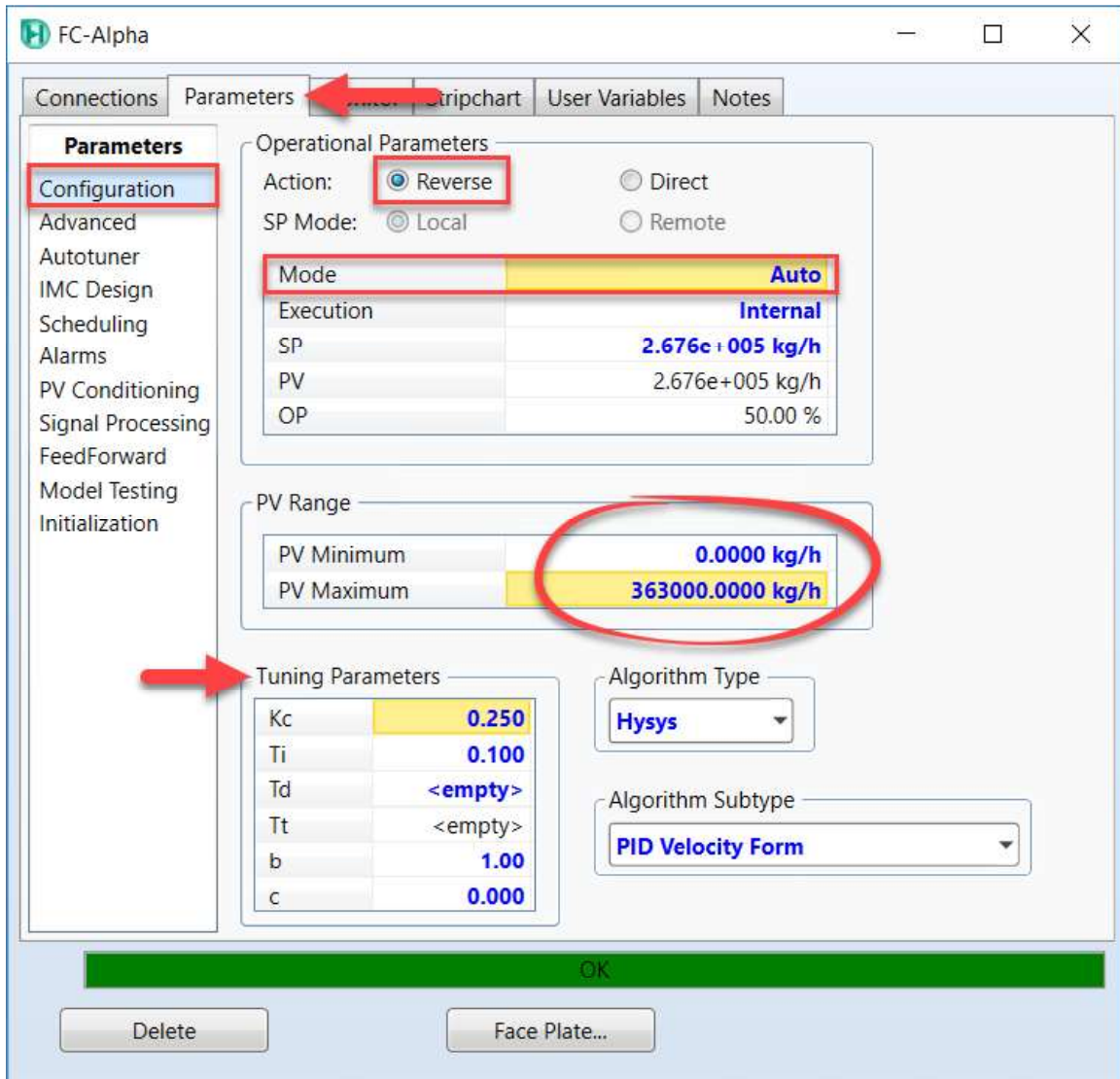
Object: VLV-100 Select OP...

Variable: Actuator Desired Position

Unknown Ranges for PV

Delete Face Plate...

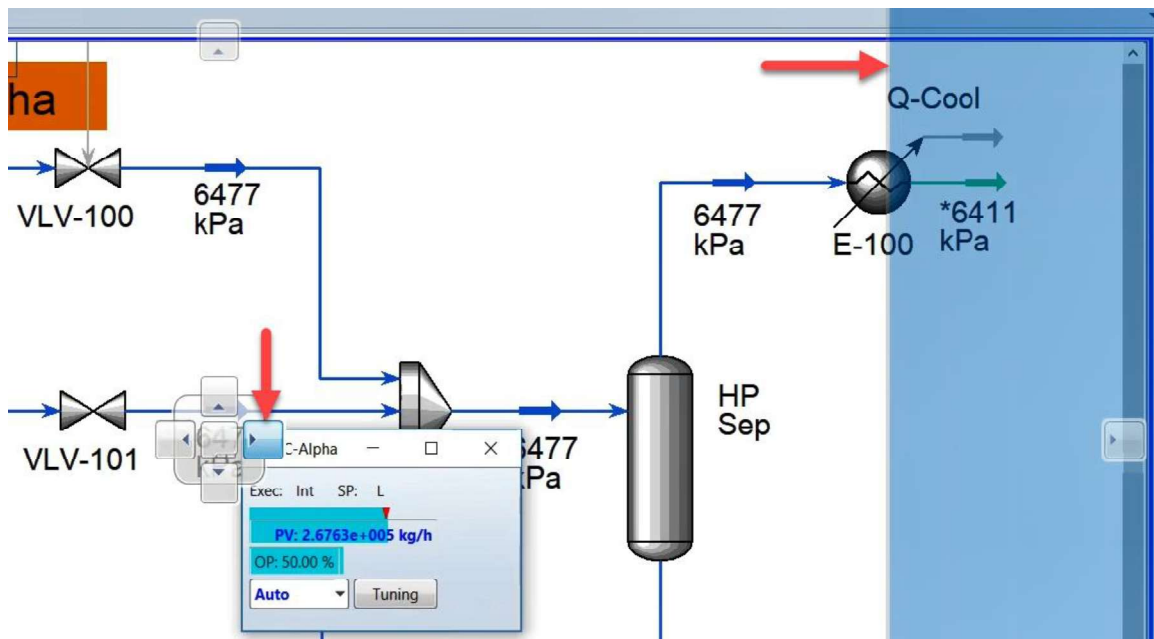
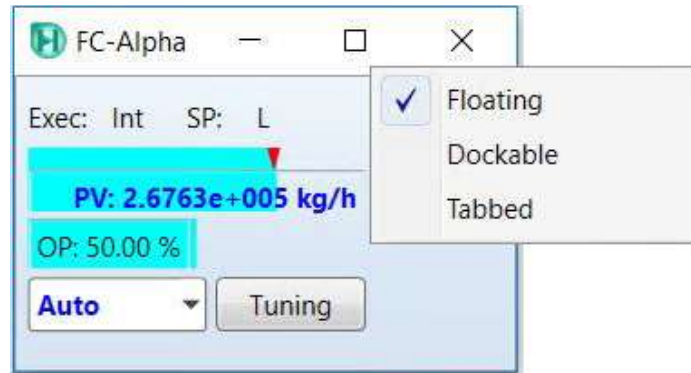
7. Go to the **Parameters | Configuration** form. Confirm that the **Action** is set as **Reverse**.
8. For the controller **PV range**, enter **0 kg/h (0 lb/hr)** for the **PV Minimum** and **363,000 kg/h (800,000 lb/hr)** for the **PV Maximum**.
9. Change the **Mode** from Manual to **Auto**.
10. Enter the following **Tuning Parameters: Kc: 0.25, Ti: 0.10 min**. The Configuration form should look like the image below:



11. Insert a Controller Face Plate for monitoring purposes by clicking on the **Face Plate** button. Place the Face Plate in a convenient location on the flowsheet.



Note: You can organize face plates by making them Dockable, right-click on them and choose this option instead of Floating (default).



12. Close the **FC-Alpha** property view window.

13. Add another **PID Controller** to the flowsheet to control the mass flow of stream Bravo. Set it up with the information below:

In This Cell...	Enter
Connections	
Controller Name	FC-Bravo
Process Variable Source	Bravo1, Mass Flow
Output Target Object	VLV-101, Actuator Desired Position

Parameters	
Action	Reverse
Range PV Minimum	0 kg/hr (0 lb/hr)
Range PV Maximum	363000 kg/hr (800000 lb/hr)
Kc	0.25
Ti	0.10 minutes
Mode	Auto

14. Generate a **Face Plate** for **FC-Bravo**. Close the property view window.

15. Add another **PID controller** to control the mass flow of stream Charlie.

In This Cell...	Enter
Connections	
Controller Name	FC-Charlie
Process Variable Source	Charlie1, Mass Flow
Output Target Object	VLV-102, Actuator Desired Position
Parameters	
Action	Reverse
Range PV Minimum	0 kg/hr (0 lb/hr)
Range PV Maximum	363000 kg/hr (800000 lb/hr)
Kc	0.25
Ti	0.10 minutes
Mode	Auto

16. Generate a **Face Plate** for **FC-Charlie**. Close the property view window.

17. Add a new **PID Controller** to control the liquid level in HP Sep.

In This Cell...	Enter
Connections	
Controller Name	LC-HP Sep
Process Variable Source	HP Sep, Liquid Percent Level
Output Target Object	VLV-103, Actuator Desired Position
Parameters	
Action	Direct

Range PV Minimum	0%
Range PV Maximum	100%
Kc	1.0
Ti	3.0 minutes
Mode	Auto
SP	30%

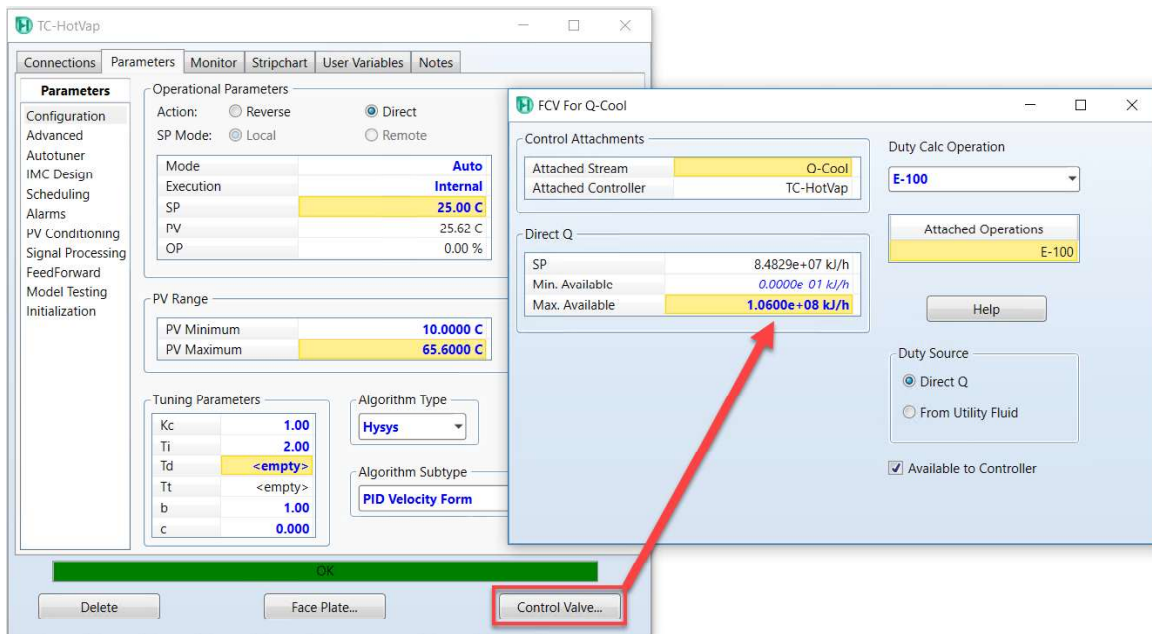
18. Generate a **Face Plate** for LC-HP Sep. Close the property view window.

19. Add another **PID Controller** to control the temperature of stream HotVap by manipulating the E-100 cooler heat duty.

In This Cell...	Enter
Connections	
Controller Name	TC-HotVap
Process Variable Source	Hot Vap, Temperature
Output Target Object	Q-Cool, Control Valve
Parameters	
Action	Direct
Range PV Minimum	10 °C (50 °F)
Range PV Maximum	65.6 °C (150 °F)
Kc	1
Ti	2 minutes
Mode	Auto
SP	25 C

20. Click on the **Control Valve** button at the bottom of the current window. For the duty stream there is no valve placed on the PFD. You will access the control valve of the duty stream by pressing the control valve button in the controller interface.

Control Valve	
Minimum Available	0 kJ/h (0 Btu/hr)
Maximum Available	1.06e+08 kJ/h (1.00e+08 Btu/hr)



21. Close the control valve window.

22. Generate a **Face Plate** for **TC-HotVap**. Close the controller property view window.

The control valve view for TC-HotVap will vary depending on the dynamic cooler model chosen (see the Dynamics | Specs page in E-100). On the Model Details form, if the Product Temp Spec radio button was selected, then there would be no need for a controller at all. Duty will float to maintain the desired temperature. If the Duty Fluid radio button was selected, then the controller would control the flowrate of the duty fluid. The user would need to enter values for UA, duty fluid C_p and the duty fluid inlet temperature.

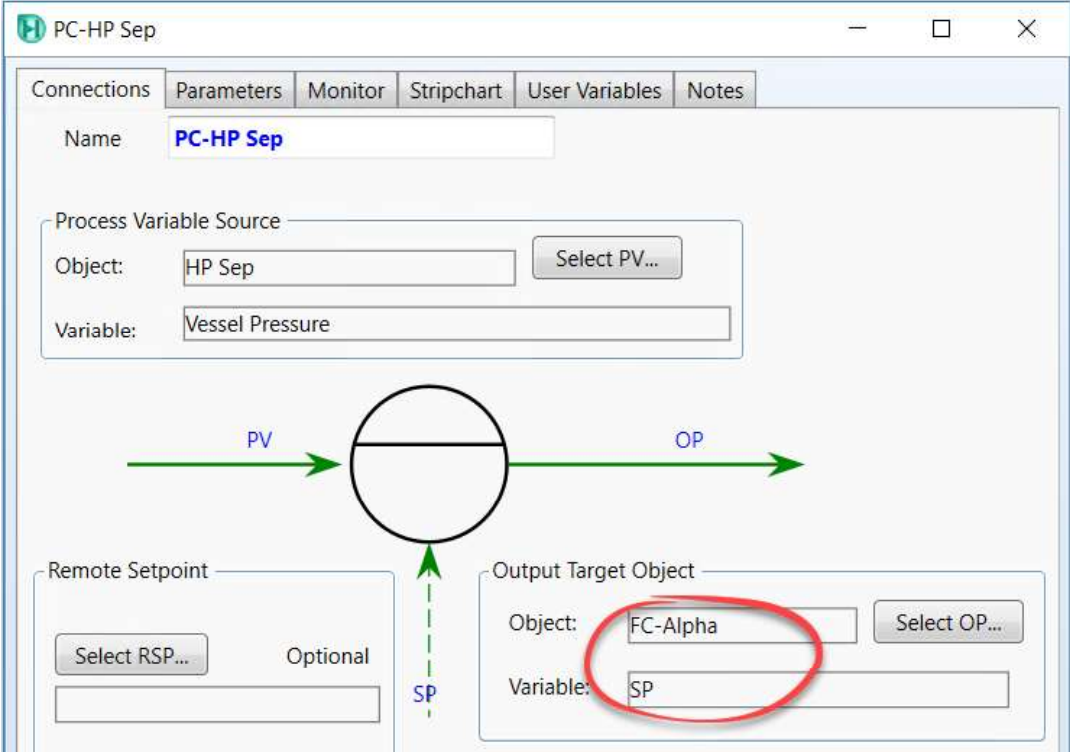
Another control option you will investigate is cascade control. Cascade control is a common control technique that uses two controllers within one feedback loop. One controller is "nested" inside the other. This means that the two controllers are not independent but linked together with the "primary" controller setting the SP for the "secondary" controller.

Cascade control can improve the dynamic response and control of a process that has considerable dead time or where the response time of the primary loop is very large.

23. Add another **PID Controller** to the flowsheet and name it as **PC-HP Sep**.

24. Set the **HP Sep | Vessel Pressure** as the **Process Variable Source** for the new controller.

25. Click on the **Select OP** button. Select the set point (SP) of controller **FC-Alpha**. Click on **Select**.



26. Open the **Parameters | Configuration** form and enter the following data:

Control Settings	
Parameters	
Action	Reverse
Range PV Minimum	5515 kPa (800 Psia)
Range PV Maximum	6895 kPa (1000 Psia)



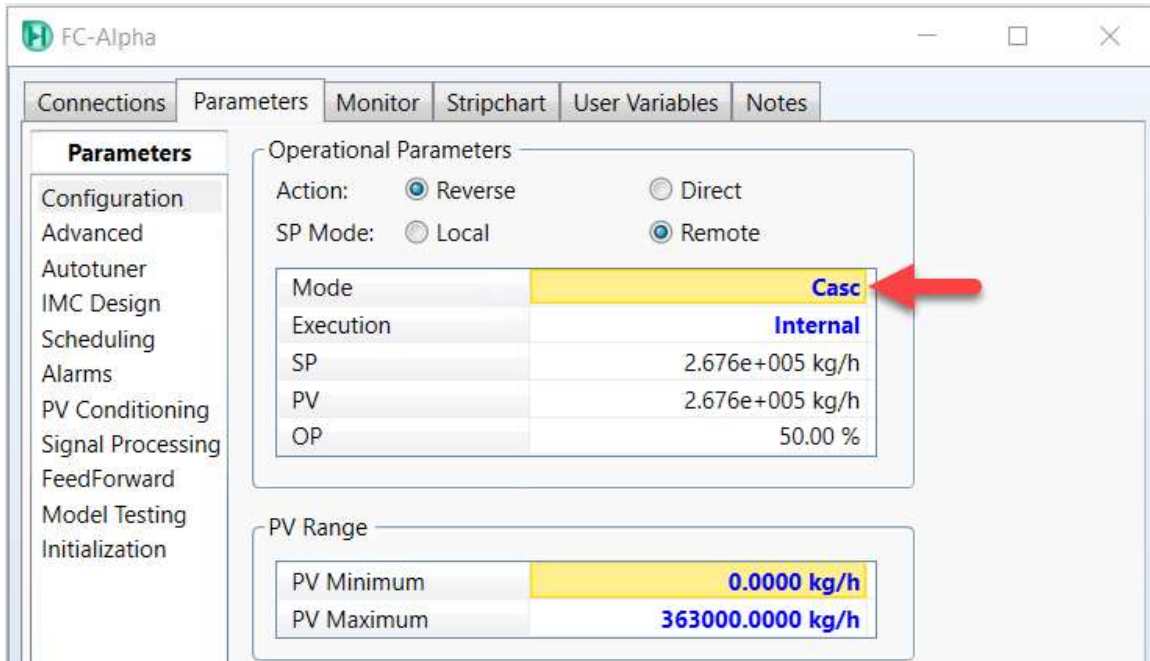
QUESTION: Has the controller operating mode changed to **Man** (manual)? [Write your answer below]

If so, enter the tuning parameters from the table below. **If not**, a mistake was made somewhere. In this case, delete the controller and create this controller from scratch again. Once the controller operating mode changes to manual, enter the following tuning parameters.

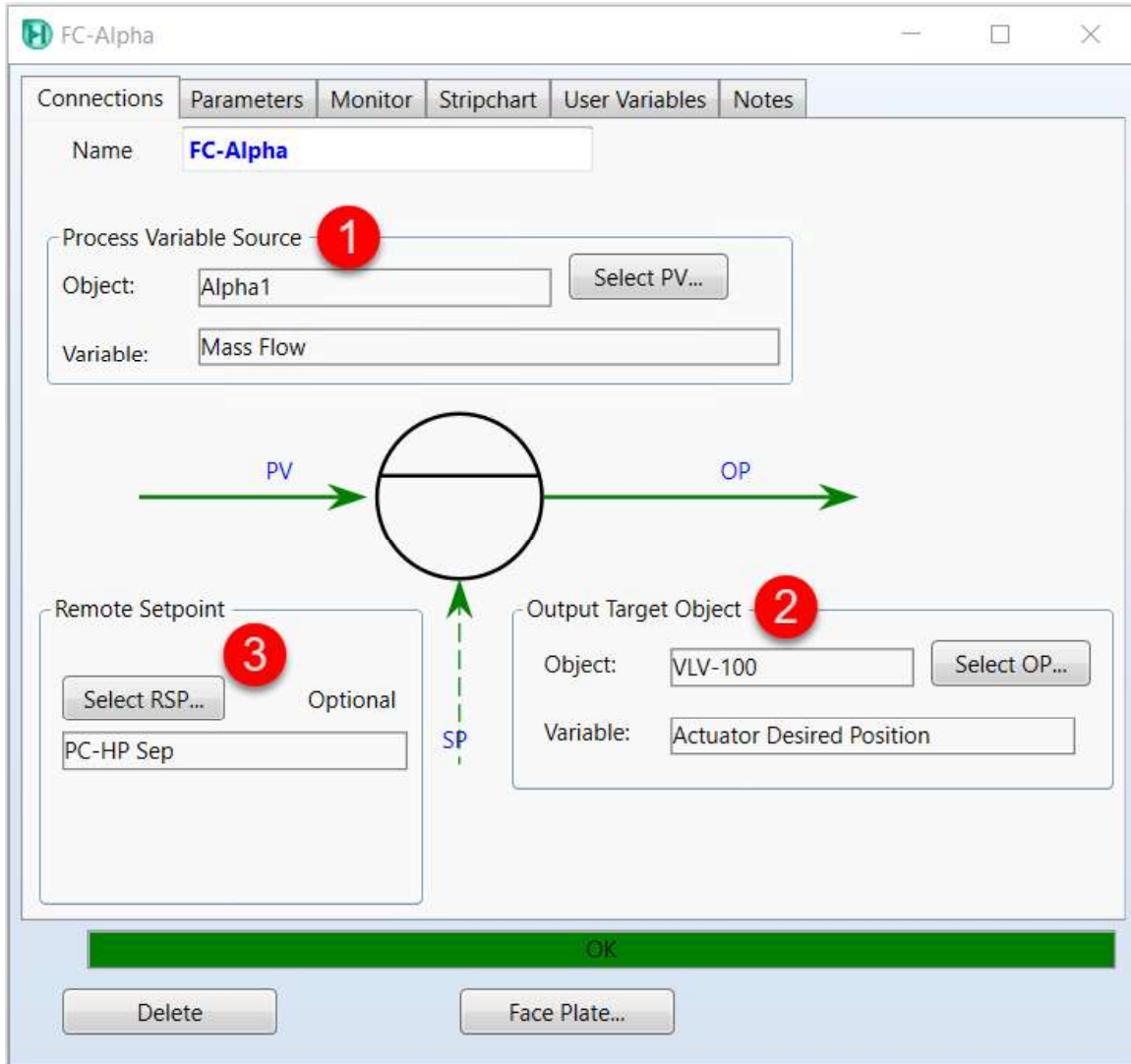
Kc	3.0
Ti	2.0 minutes
Mode	Auto

27. Generate a **Face Plate** for **PC-HPSep**.

28. Open **FC-Alpha** and change its operating mode to **Casc** (cascade).



29. Go to the **Connections** form of the **FC-Alpha** controller. Now, it should have three connections. Note that this is the first time that more than two connections have been made for a controller and it is typically done for cascade control strategies.



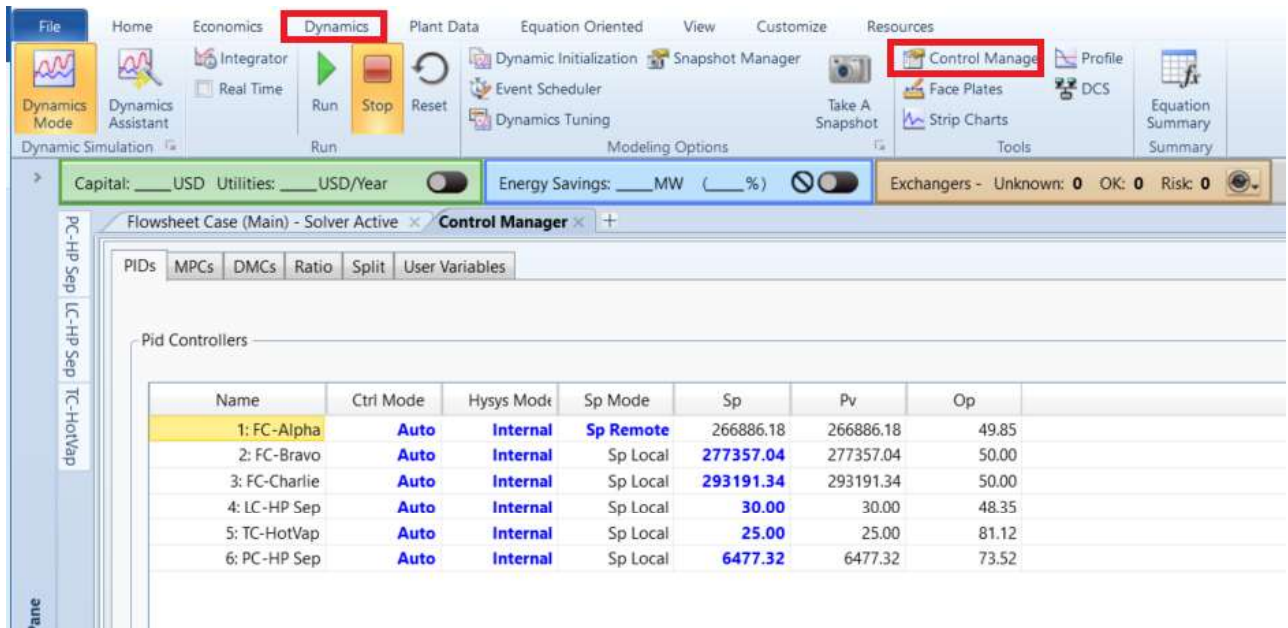
30. Close the **FC-Alpha** property view window.

31. Run the model for about a minute. The model should reach steady-state conditions quickly.

32. **Stop** the dynamic run.



Note: Use face plates to check how the model steadies. You can open the Control Manager to check how controllers are performing. If any of the controllers' output (OP) is close to either 0 or 100, then either the controller action is wrong or the control valves are not correctly sized.



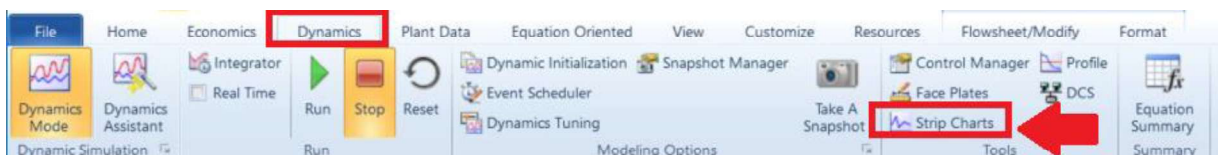
33. Save your file as **02-DynamicWithControllers.hsc**.

Task 2 – Build Strip Charts and Observe the Dynamic Behavior of the Model

While the model runs in dynamic mode, it is difficult to observe the dynamic behavior of process variables. Some process variables can be monitored by looking at either a property view window or the Workbook. All process variables will be updated as the model runs. Using a different reporting tool called Strip Chart allows users to monitor several variables in real-time as the dynamic model runs.

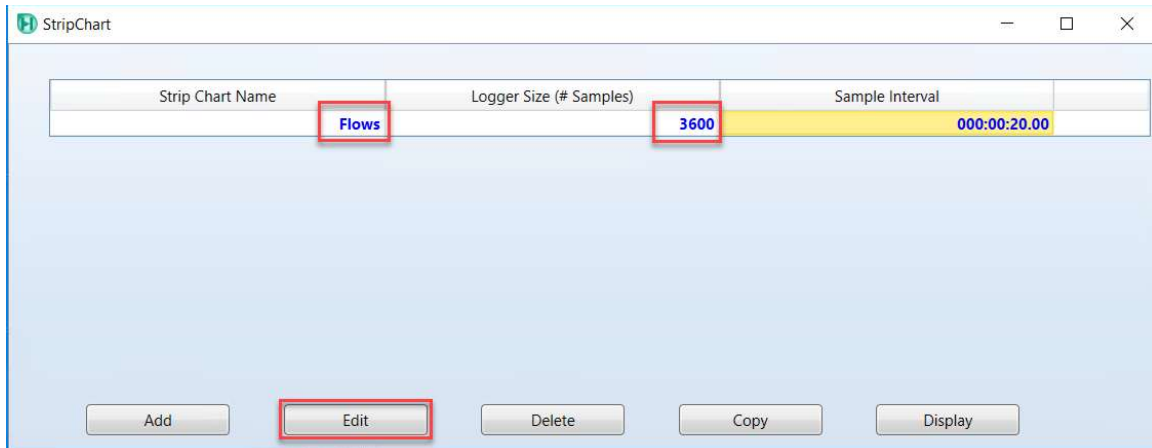
Strip charts provide a method for easily tracking key process variables in a graphical environment. Strip charts are built individually. Multiple strip charts are allowed, and each strip chart can have an unlimited number of variables charted; though it is wise to limit each chart to up to six variables. Follow the steps below to create a couple of strip charts to monitor key process variables.

34. Go to the **Dynamics** tab on the ribbon and click on the **Strip Charts** button.

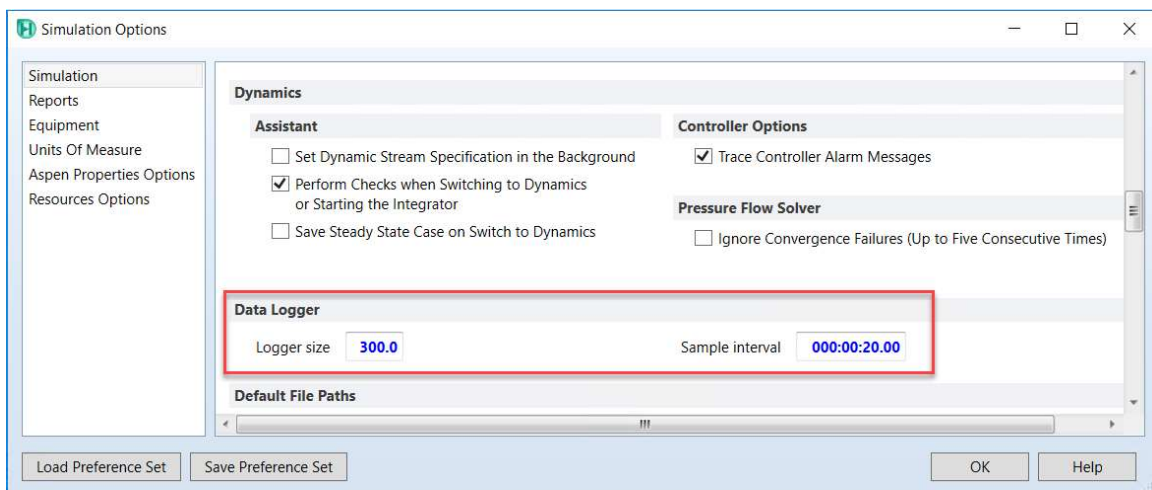


35. In the **Strip Chart** window, click on the **Add** button. A new strip chart titled as DataLogger1 (default name) will be created. Change its **name** to **Flows**.

36. Set the Logger Size to 3600 and keep the Sample Interval at 20 seconds.

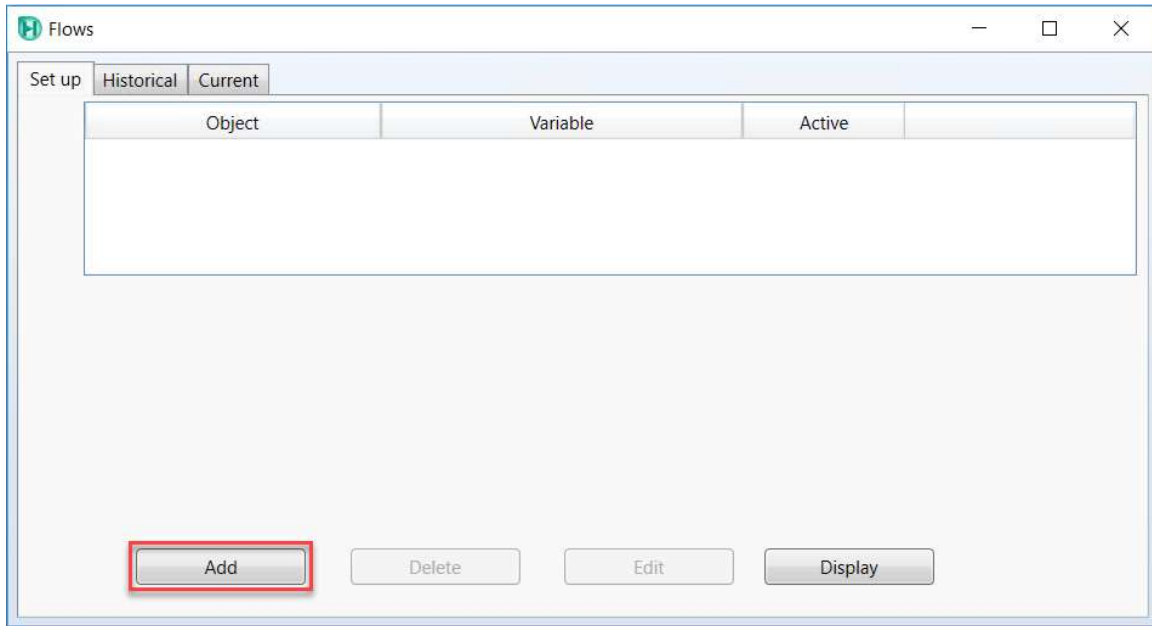


Note: You can change the default logger size and sample interval in the simulation options (File | Options).



37. Click on the **Edit** button to go to the strip chart setup form.

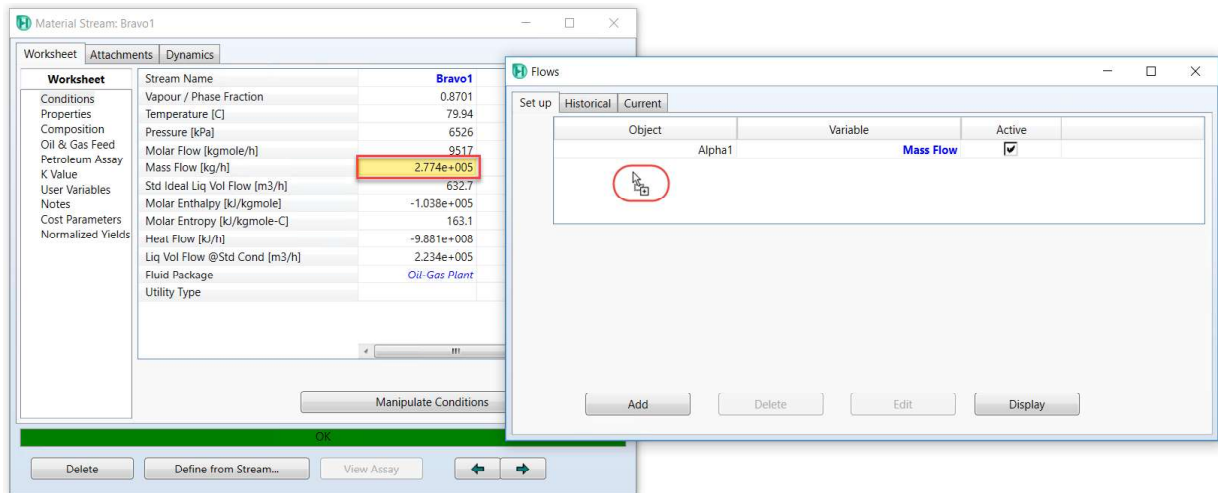
38. Click on the **Add** button to add variables to monitor to the strip chart.



39. Add Alpha1 | Mass Flow.

40. You can add a variable by dragging and dropping it to this form. **Double click** on stream **Bravo1**. Keep Bravo1 and the strip chart setup windows side by side.

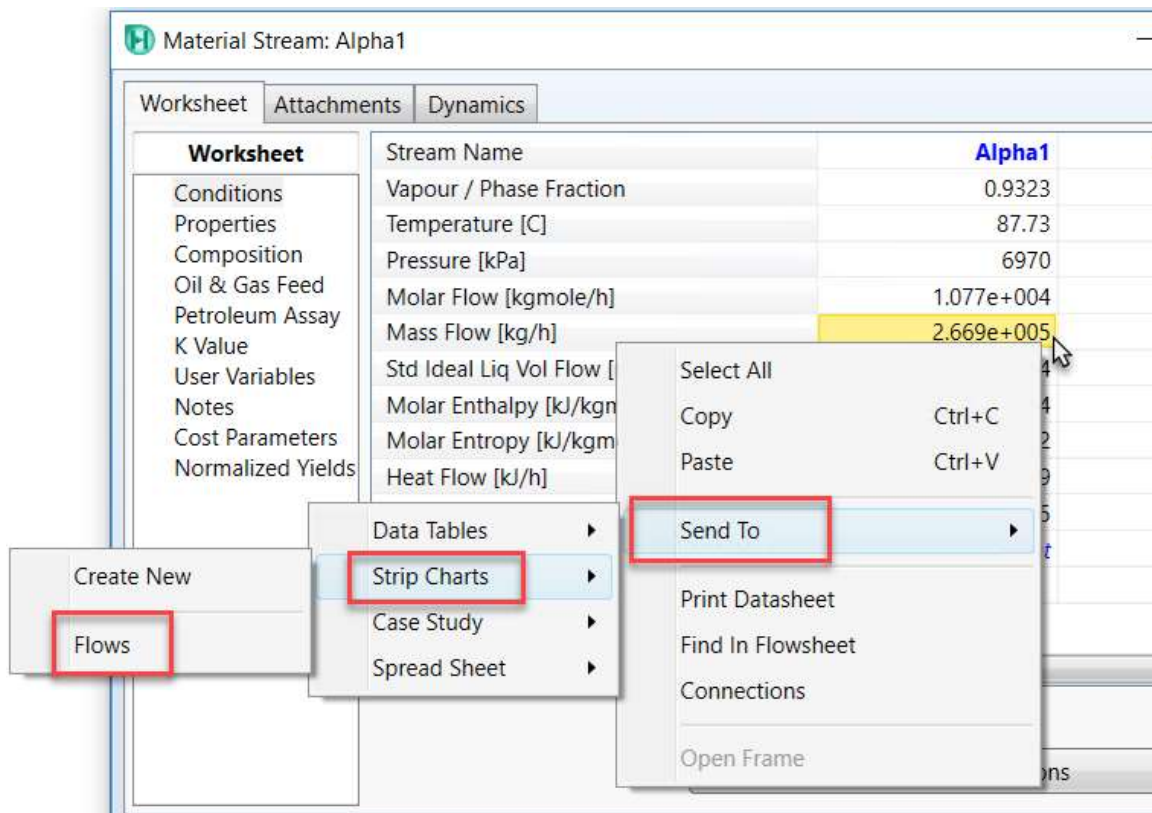
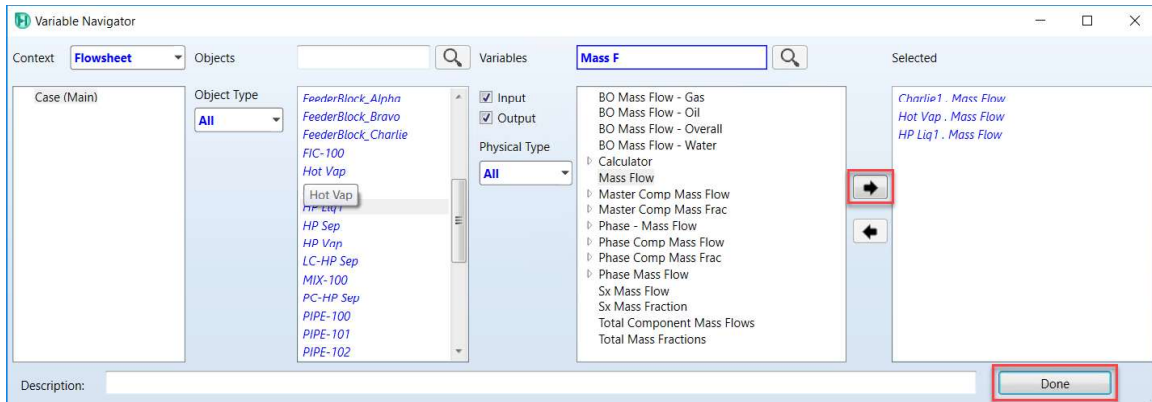
Left click on the **mass flow** of stream **Bravo1**. Hold it. Then **drag and drop** it into the strip chart setup window shown above.

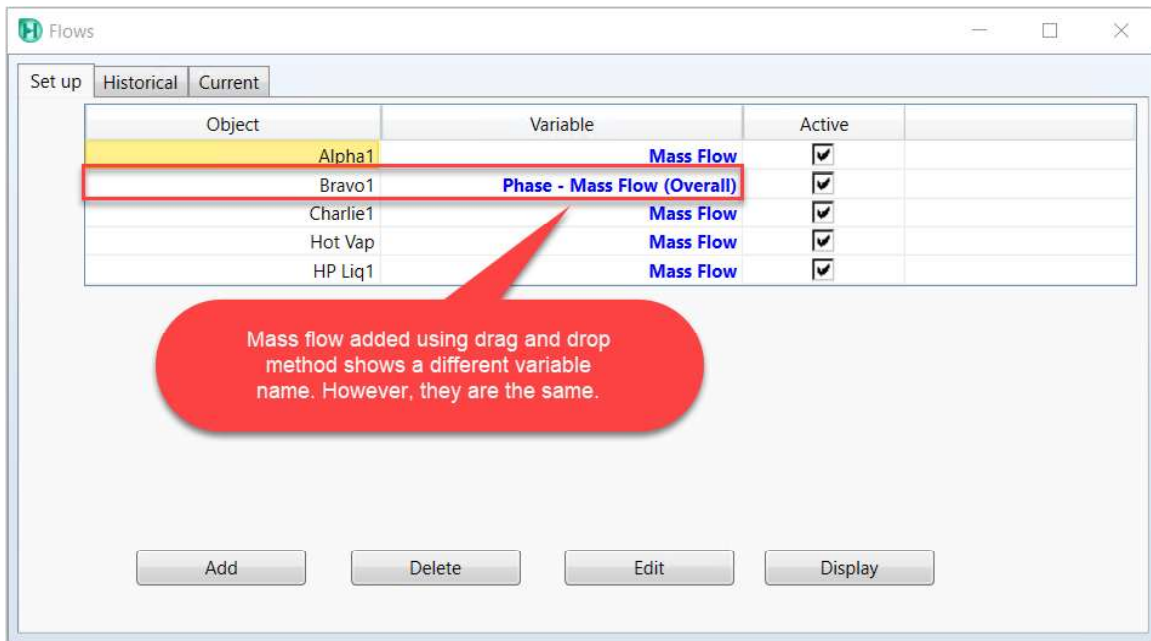


41. Add the following variables using either the Variable Navigator or the drag and drop method.

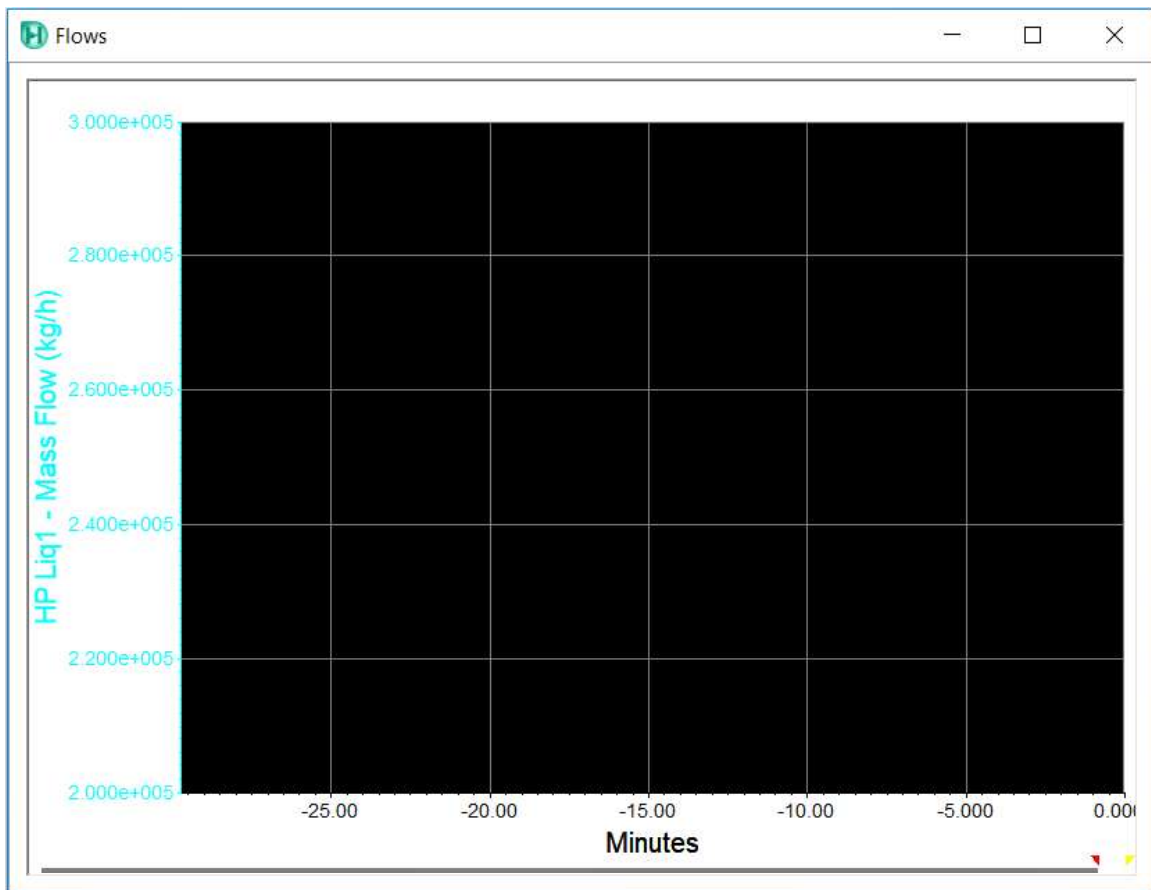
- Charlie1 | Mass Flow
- HotVap | Mass Flow
- HPLiq1 | Mass Flow

If you use the Variable Navigator, after finding a variable, just use the contextual **Arrows** to add the selected variables to the strip chart. After adding all the necessary variables, click on **Done** to close the Variable Navigator.

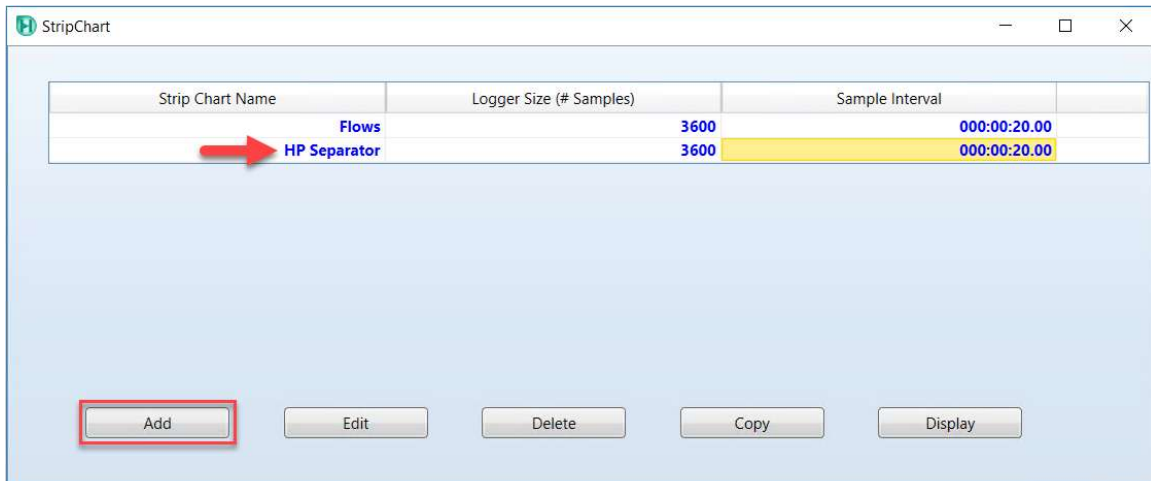




42. Click on the **Display** button to launch the Flows strip chart.



43. Close the Flows strip chart setup window and jump back to the main strip chart menu.
44. Click on the **Add** button to add a second strip chart. Rename it as **HP Separator** with the same logger size and sample interval settings as those of the Flows strip chart.



45. Click on the **Edit** button and add the following variables:

- HP Sep | Vessel Pressure
- HP Sep | Liquid Percent Level
- Hot Vap | Temperature

46. **Display** the HP Separator strip chart.

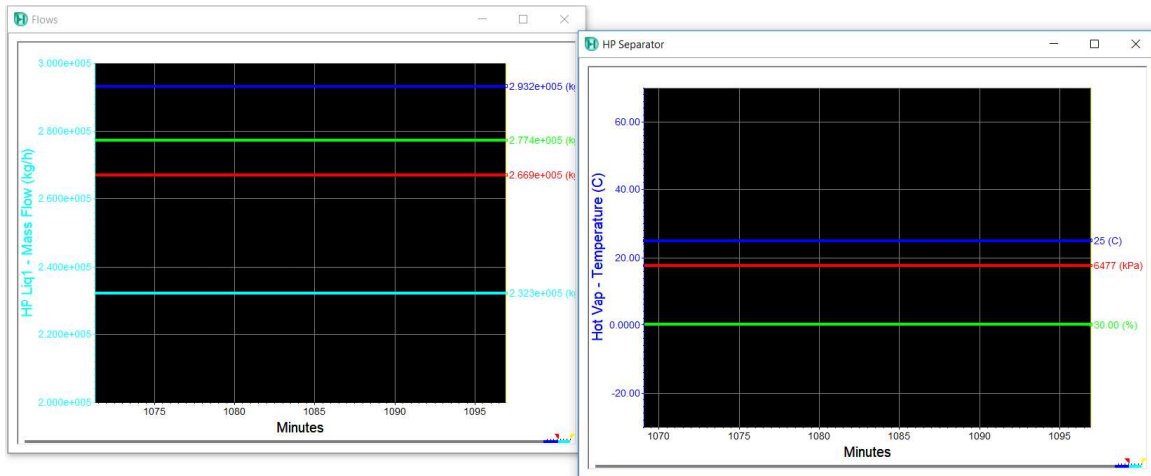
47. Save your case as **02-DynamicWithControllersStripCharts.hsc**.

When running the model in dynamic mode, it can be difficult to monitor the behavior and status of simulation variables as they change over time. So, reporting and verifying results is different from steady-state. That is why the controller face plates and strip charts were considered. Place these in a convenient place on the flowsheet so that you can easily track results.

48. **Run** the model in dynamic mode and check the Flows strip chart.



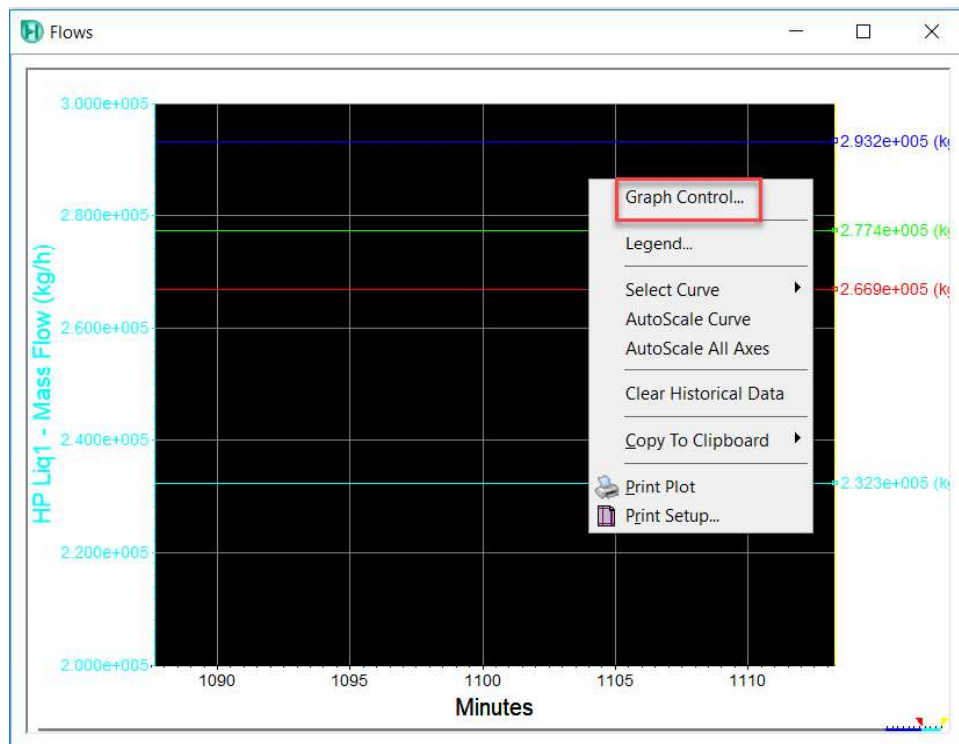
QUESTION: Has steady-state been attained? [Write your answer below]



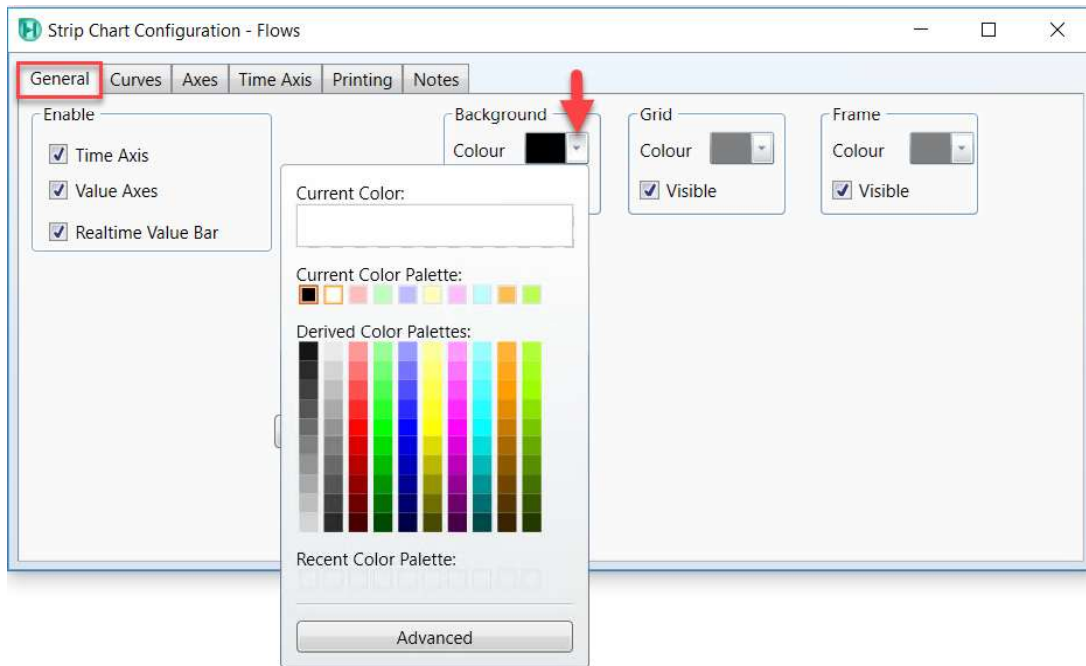
49. **Stop** the Integrator and **Save** your case.

Task 3 – Customize Strip Charts and Change the Controllers Set Point

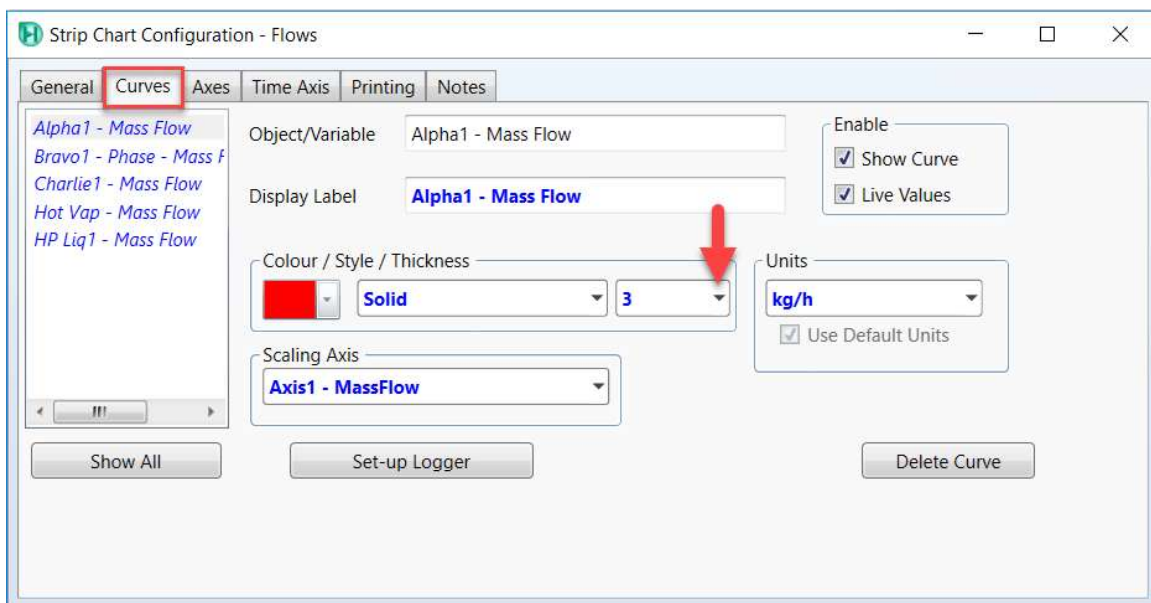
50. **Right-click** on the Flows strip chart. Click on the **Graph Control** option. It has several tabs to customize color, curves, axis, print options, etc.



51. Change the **background** color to **white**.



52. Go to the **Curves** tab. Select **Alpha1- Mass Flow**. Change the line thickness to **3**.



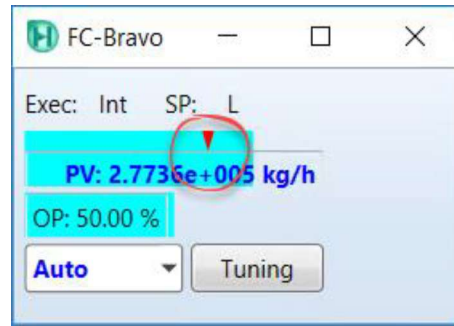
53. Change the line thickness for the rest of the curves to **3**. Close the graph control window.

54. Open the graph control form for the other strip chart. Change the background color to **white** and change the line thickness for all the curves to **3** as well.

55. Save the file as **02-DynamicWithControllersStripChartsCustom.hsc**

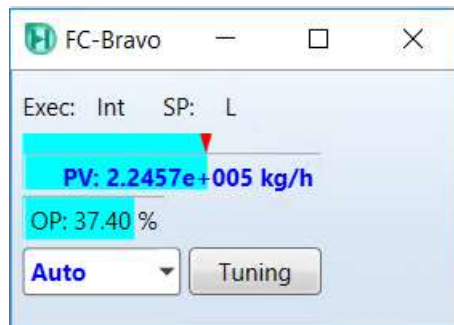
There are two ways in which the set point of a controller can be changed using a face plate – using either the sliding bar or entering a numerical value.

56. Hold the **red down arrow** of the **FC-Bravo** face plate and move it towards the left to reduce the set point to any arbitrary number.



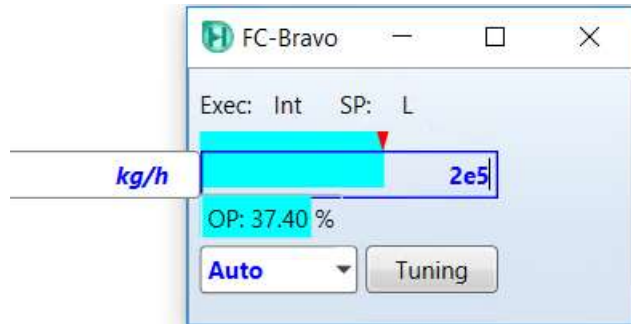
57. Run the integrator. Observe the changes occurring in both the **Flows** strip chart and **FC-Bravo** face plate. Slowly the flow, which is indicated by a number and light blue area, will come to the new set point.

58. Stop the solver after a few seconds.

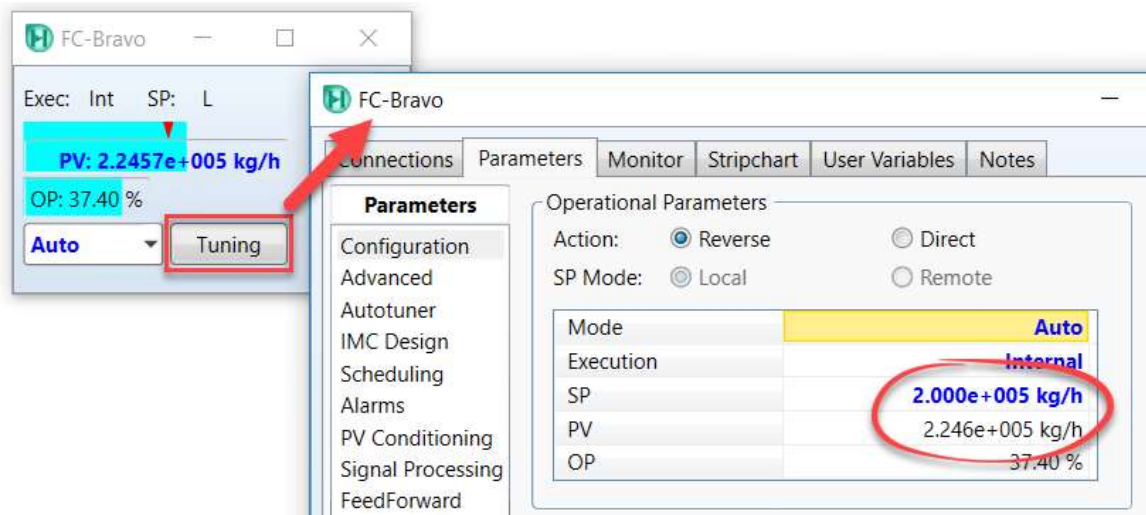


Now, you will change the set point by typing in a numerical value. The PV value in the face plate shows the current PV (process variable) value, which is flowrate. However, if you type a new number, it will become the new set point. The new entered number will be found in the controller form but not in the face plate. The sliding bar will indicate the change though.

59. Click on the **PV** cell of the FC-Bravo face plate; left-click on the current number and type in **2e5 kg/h (4.4e5 lb/h)**. Notice that the red sliding bar changed its position to indicate the new set point.

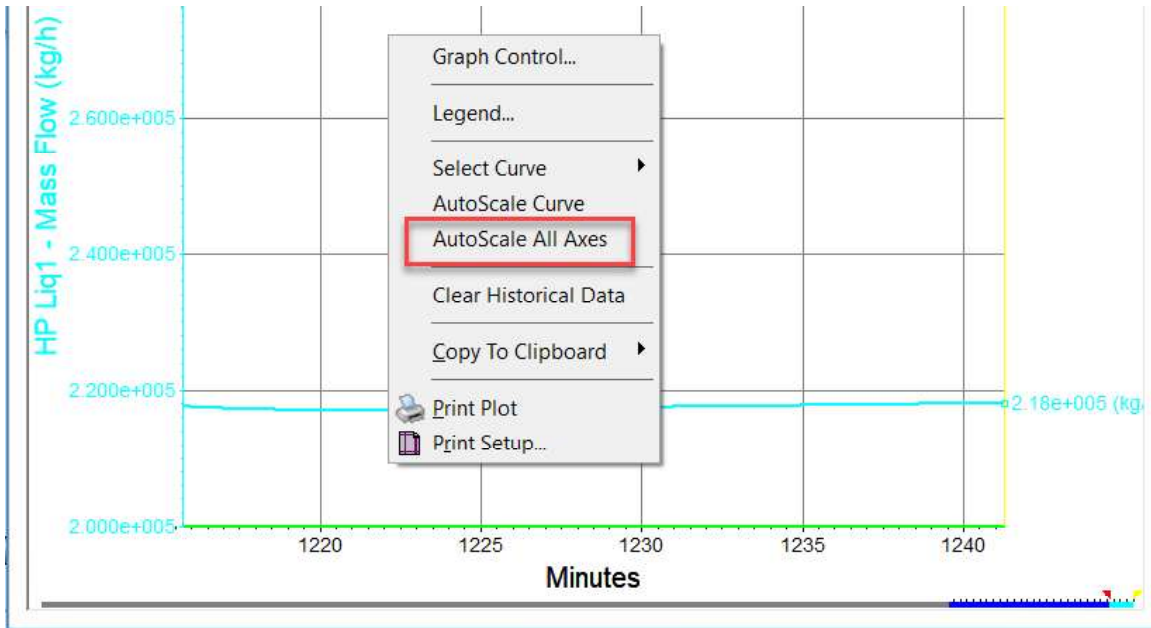


60. Click on the **Tuning** button to launch the controller form. Notice that the set point has changed.



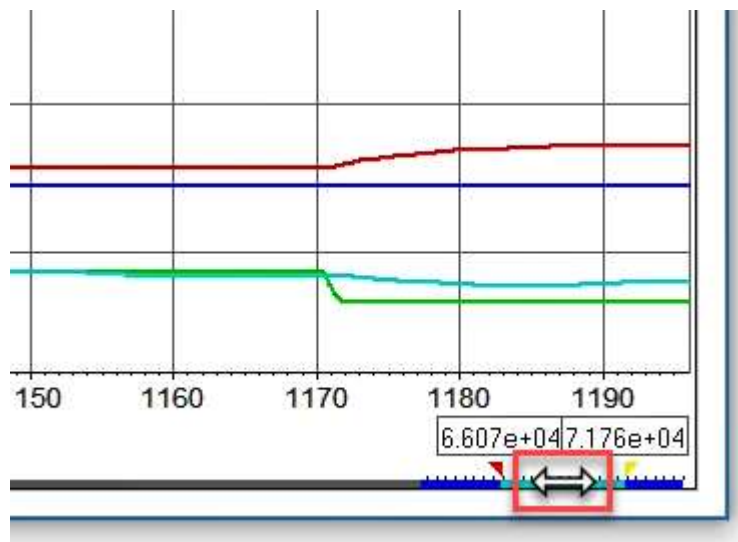
61. **Run** the integrator for a few seconds and then **stop it**.

62. **Right-click** on the flows strip chart and select the **Auto Scale All Axes** option. It will automatically adjust the scales of all curves.





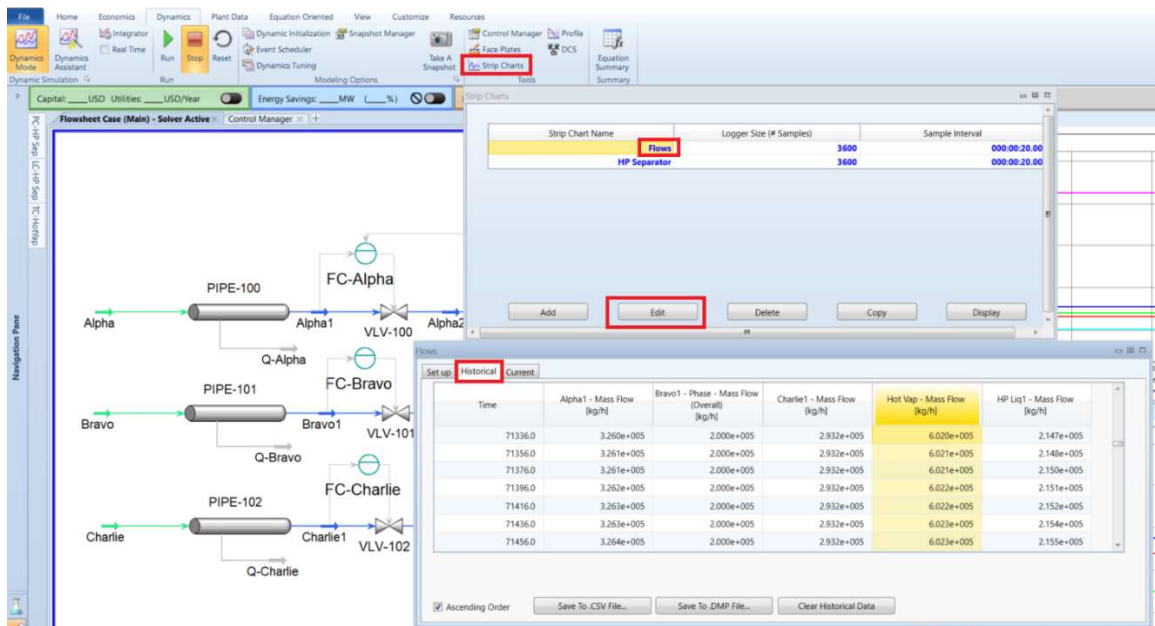
63. Click and hold somewhere **in between** the red and yellow triangle and move it to the left or right to zoom into a time zone keeping the same time span.



64. Change the **set point** of controller **FC-Bravo** back to **2.77e5 kg/h (6.1e5 lb/h)**.
Run the model just for a few seconds.
65. Drag the **yellow sliding triangle** all the way to the right side.
66. Stop the integrator.

Now that the dynamic run has been completed, it would be interesting to look at the historical data of key process variables that we could possibly be interested in. That can be easily done with strip charts. Let's assume we want to look at the Hot Vap – Mass Flow historical data.

67. Click on the **Strip Charts** button. Click on the chart which contains the variable(s) you are interested in. Click on the **Edit** button and go to the **Historical** tab of the strip chart. The historical data can be exported to **MS Excel** as a ***.csv** file (or copy-paste the data into an Excel ***.xlsx** / ***.xlsm** file) or as a ***.dmp** file. See the image below:

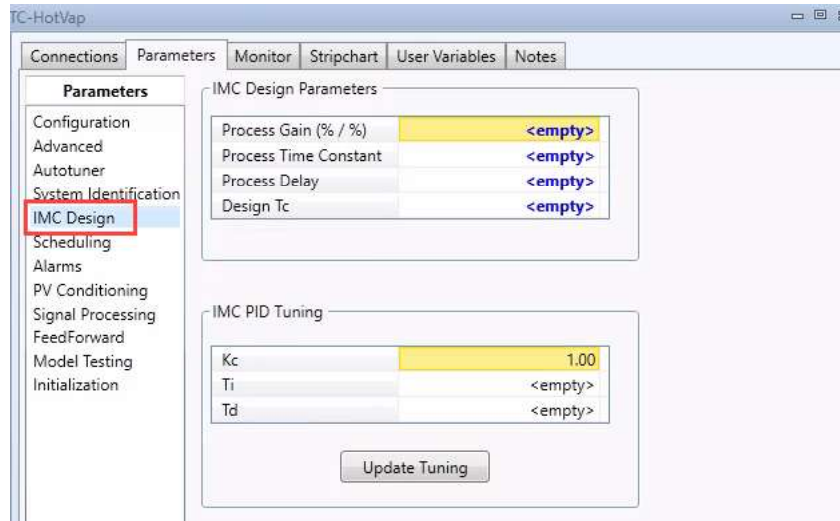


68. Save your file

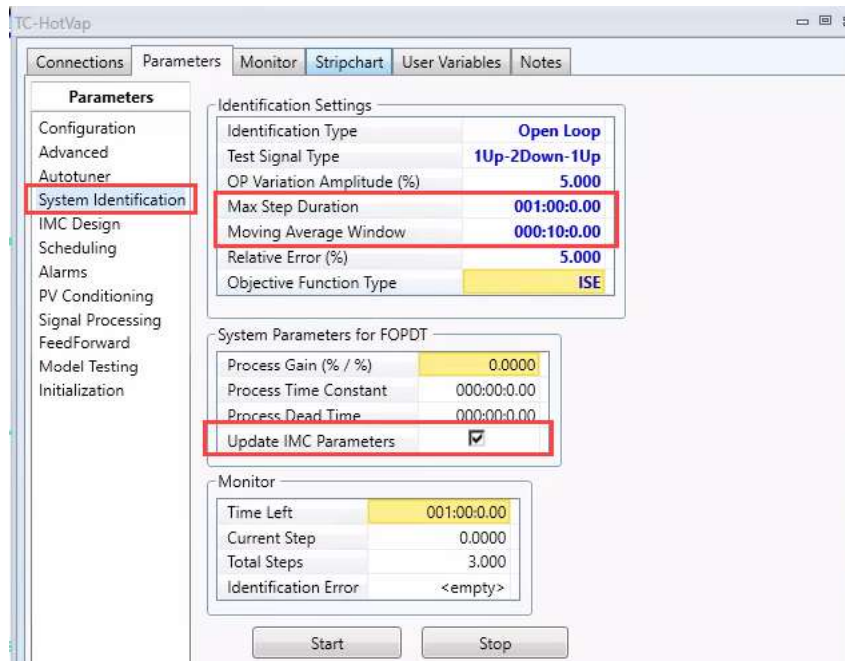
Task 4 – Estimate PID Controller Tuning Parameters (Optional)

A new feature was released in Aspen HYSYS Dynamics V14 to estimate the tuning parameters for a PID controller. On the System Configuration page, the parameters for a First Order model Plus Dead Time model (FOPDT) are calculated based on the results of a step test. These parameters can be used on the Internal Mode Control (IMC) Design page to calculate the PID parameters. The IMC method is common in most of the process industries and has a solid theoretical basis.

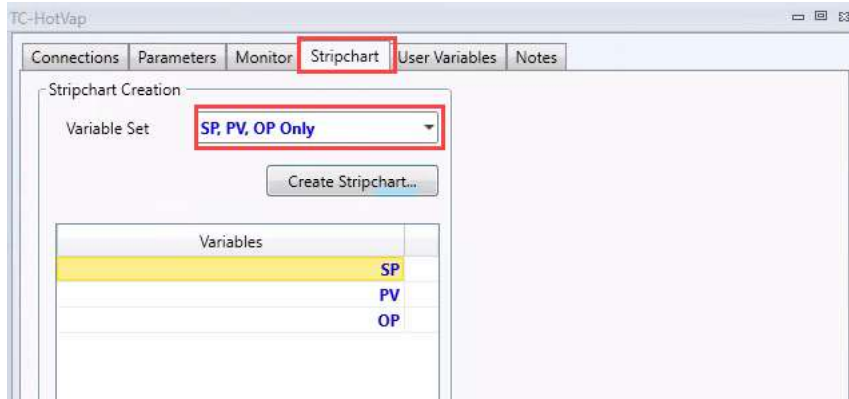
69. Go to the **Parameters | IMC Design** form of the **TC-HotVap** controller and observe that the parameters are empty.



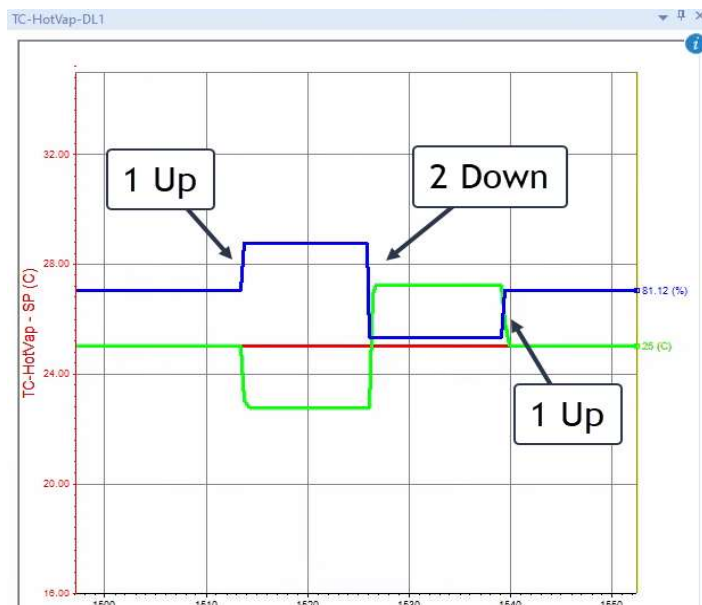
70. Go to the **Parameters | System Identification** form. Set the **Max Step Duration** to 1 hour (3600 seconds) and the **Moving Average Window** to 10 minutes (600 seconds). Verify that the **Update IMC Parameters** checkbox is enabled. Leave the other values as their default settings.



71. On the **Stripchart** tab, select the “**SP, PV, OP Only**” variable set and click **Create Stripchart**. Some objects in HYSYS Dynamics, such as PID controllers, have a Stripchart tab to quickly create pre-defined stripcharts.



72. Display the strip chart and edit the Graph Control settings based on your preferences. Select Autoscale All Axes.
73. Run the model for a short time to populate the strip chart. Then stop the integrator.
74. On the **Parameters | System Identification** form of the **TC-HotVap** controller press Start. After a short time the step test should complete. The strip chart should look like the image below with the 1Up-2Down-1Up test.



75. Go to the **IMC Design** form and observe the IMC PID Tuning results.
76. Save the file as **02-DynamicWithControllersPIDTuning.hsc**

Compressor Curves and Surge Control

Workshop

Files can be found in folder:

C:_Workshops\EHY\EHY223\EHY223.12.01\04_CompressorSurge on your cloudshare environment

Objective

In this workshop, compressor curves will be used to model the behavior of a high-pressure gas compressor. Using performance and efficiency curves to model rotating equipment allows Aspen HYSYS Dynamics to accurately simulate the dynamic behavior of plant equipment. You will also learn how to implement a compressor surge control strategy to protect the equipment against a potentially dangerous low-flow operating scenario.

Description

Using compressor curves for rotating equipment in an Aspen HYSYS Dynamics model allows you to accurately simulate existing compressors. You also have the same capability for pump models. With a robust model, you can determine whether an existing compressor (or pump) is able to meet the operating specifications of the process.

Compressor performance curves also allow Aspen HYSYS Dynamics to compute heads and efficiencies that depend on flowrate. This results in improved accuracy and a better chance to rigorously model existing plant equipment.

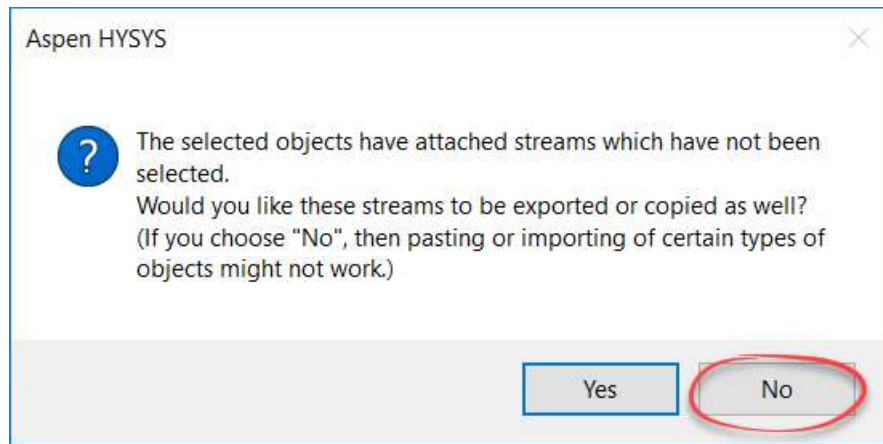
This workshop includes the following tasks:

- Task 1 – Import the Compressor Model.
- Task 2 – Set up the Compressor.
- Task 3 – Add the Surge Controller.

Task 1 – Import the Compressor Model

In this task, you will add a compressor into your already operating dynamic model.

1. Included with the course workshop files, you will find a starter file entitled **04-CompCurves-Starter.hsc**. This file contains a compressor using a series of compressor performance curves.
2. Open 04-CompCurves-Starter.hsc.
3. Right-click on the **Stage 1** compressor on the flowsheet and select **Copy**.
4. Click on **No** when prompted to copy attached streams information along with the compressor.



5. Close the starter case without saving any changes.
6. Open the file saved **03-ExpandModel5_AfterRelief.hsc**.
7. Right-click on the background of the flowsheet and select **Paste** to insert the copied compressor.



Note: You might get the following message:

Aspen HYSYS
✕

?

The changes you made to Oil&Gas Plant may result in different enthalpies and other properties. Since dynamics continues from an existing solution, this can cause unrecoverable flash problems or kicks that will slow down the simulation.

This can be handled by updating operations to reflash all affected streams and re-initialize the next time the integrator is started. This update will cause a kick in the simulation, but it should be able to recover.

However, if your changes should not have a significant impact (e.g. you only added a component or removed one that had zero composition everywhere) you may skip the update.

Do you wish to update affected operations?

Since you have pasted a compressor model from a steady-state simulation ('04-CompCurves-Starter.hsc') into a dynamic model ('03-ExpandModel5_AfterRelief.hsc'), this 'warning' message is basically telling you that there might be convergence issues due to flash calculations. This is because the compressor and its related streams data might not be consistent with the existing dynamic model. You will enter the missing data for the compressor to run in dynamic mode later, and 'initialize' this unit by running the integrator for a short while. Click on the 'Yes' button.

8. Save the file as **04-Comp.hsc**.

Task 2 – Set up the Compressor

In this task, you will enter all the required data to fully define the compressor model to be run in dynamic mode.

9. Connect the recently added compressor model (named as **Stage 1**) to the **KO Vap** stream, which is the vapor product stream coming out of the **Knockout Drum** separator.
10. Create a compressor outlet stream named as **Comp Out** and an energy stream named as **Comp-HP**.
11. On the **Rating | Curves** page of **Stage 1**, notice that several curves are checked as active and the **Enable Curves** checkbox is also checked. On the **Dynamics | Specs** page, notice that the **Use Characteristic Curves** option is checked and the **Speed** is specified as **5000 rpm**.

Compressor: Stage 1

Design Rating Worksheet Performance Dynamics

Rating

Curves
Flow Limits
Nozzles
Inertia
Electric Motor

Efficiency
☒ Adiabatic ☐ Polytropic

Curve Name	Activate
2488 for Stage 1	☒
3733 for Stage 1	☒
4977 for Stage 1	☒
6177 for Stage 1	☒

Off-Design Correction
☐ On
 Ref. Temperature: 24.84
 Ref. Molec. Weight: 20.34

Offset Values
 Head: 0.0000 m
 Efficiency (%): 0.00

Compressor Speed
 5000 rpm

☒ Enable Curves
 View Curve...
 Add Curve...
 Delete Curve
 Clone Curve
 Plot Curves...

☐ Plot All Collections

Extrapolation
☒ Linear ☐ Quadratic

Wegstein Solver
☐ Enable acceleration bounds
 Lower bound: -5.0
 Upper bound: 0.0

Delete Ready ☒ On ☐ Ignored

Compressor: Stage 1

Design Rating Worksheet Performance Dynamics

Dynamics

Specs
Holdup
Stripchart

Dynamic Specifications

Duty [kJ/h]	<empty>	☐
Adiabatic Efficiency	84	☐
Polytropic Efficiency	84	☐
Pressure Increase [kPa]	<empty>	☐
Head [m]	2598	☐
Fluid Head [kJ/kg]	25.48	☐
Capacity [ACT_m3/h]	8899	☐
Speed [rpm]	5000	☒
Linker Power Loss [kJ/h]	<empty>	☐

☐ Reciprocating (Positive Displacement) Calculate Head

☐ Screw Compressor

☒ Use Characteristic Curves Create Surge Controller...

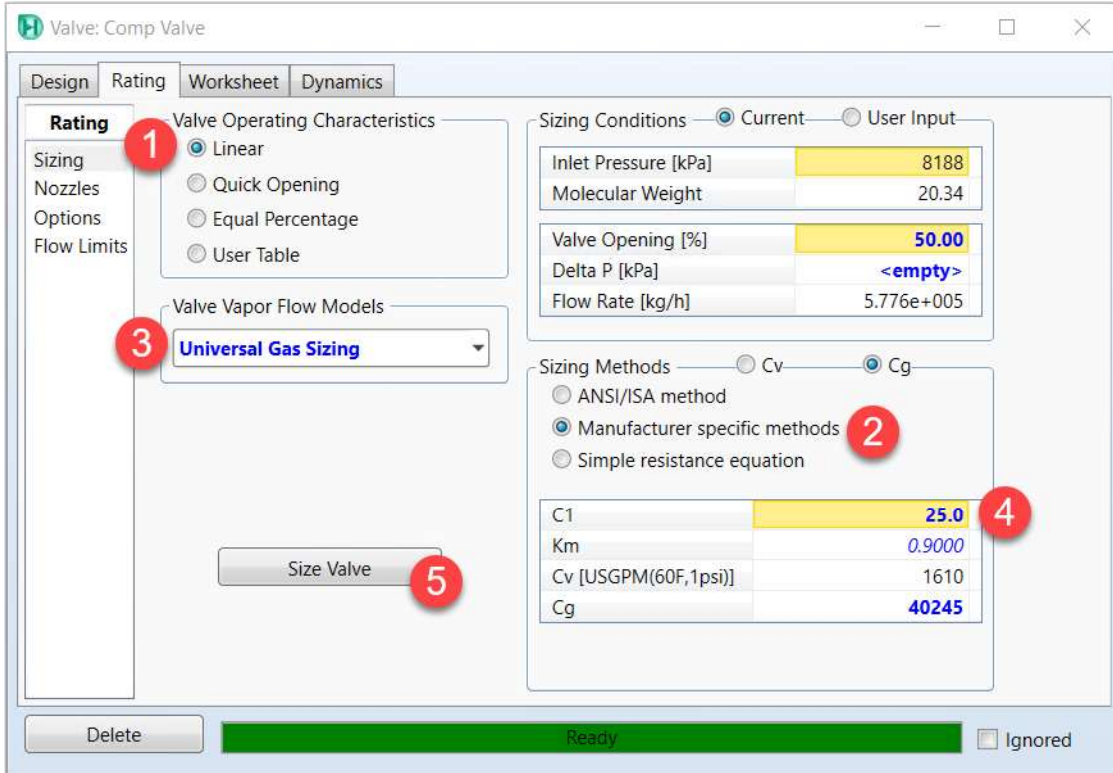
☐ Electric Motor

12. Run the Integrator for a short while to propagate information to stream **Comp Out**. Again, if the Dynamics Assistant pops up, do not make any changes.



QUESTION: What is the compressor discharge pressure? [Write your answer below]

13. Save your case as **04-Comp1.hsc** before continuing.
14. Connect a **Valve** to the compressor outlet. Name the valve as **Comp Valve** and attach an outlet stream to it, naming it as **To TEG Tower**.
15. Remove the pressure specification from the compressor suction and move it to the outlet of the discharge valve. Set the **dynamic pressure spec** as **6895 kPa (1000 psia)**.
16. Size the **Comp Valve** using the information below. Enter the data on the **Rating | Sizing** page.



Valve: Comp Valve

Design | **Rating** | Worksheet | Dynamics

Rating

Valve Operating Characteristics

- ☒ Linear
- ☐ Quick Opening
- ☐ Equal Percentage
- ☐ User Table

Valve Vapor Flow Models

Universal Gas Sizing

Sizing Conditions — ☒ Current — ☐ User Input

Inlet Pressure [kPa]	8188
Molecular Weight	20.34
Valve Opening [%]	50.00
Delta P [kPa]	<empty>
Flow Rate [kg/h]	5.776e+005

Sizing Methods — ☐ Cv — ☒ Cg

- ☐ ANSI/ISA method
- ☒ Manufacturer specific methods
- ☐ Simple resistance equation

C1	25.0
Km	0.9000
Cv [USGPM(60F,1psi)]	1610
Cg	40245

Size Valve

Delete Ready Ignored



QUESTION: What is the computed C_g for Comp Valve? [Write your answer below]

17. Add a PID Controller with the following data to fix the pressure set point at the same value observed after running the dynamic model just after you installed the compressor.

Controller Settings	
Connections	
Controller Name	KO PC
Process Variable Source	Comp Out, Pressure
Output Target Object	Comp Valve, Actuator Desired Position
Parameters	
Action	Direct
Range PV Minimum	6000 kPa (870 psia)
Range PV Maximum	10480 kPa (1520 psia)
Kc	2
Ti	2 minutes

18. Generate a **Face Plate** for the new controller.
19. **Run** the Integrator until steady-state is attained. Note that you may wish to start the pressure controller in **Manual** operating mode and then switch to Automatic mode to maintain the same discharge pressure you observed after adding the compressor.
20. Save your case as **04-CompControl.hsc**.

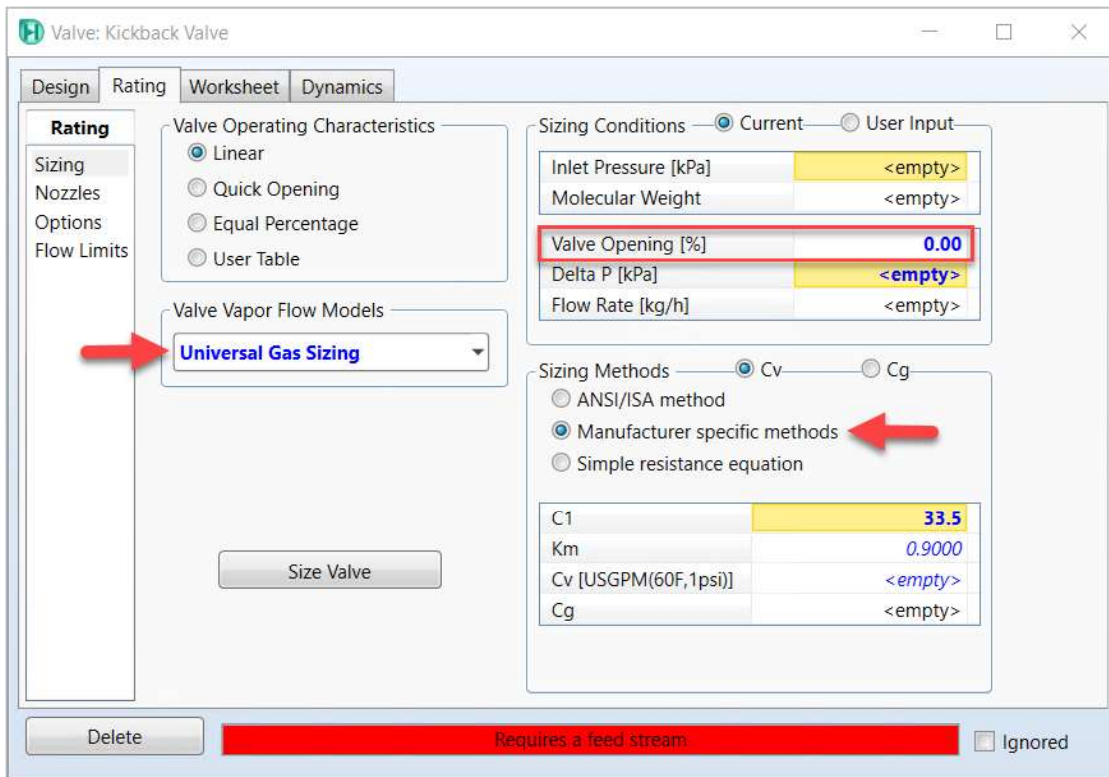
Task 3 – Add the Surge Controller

If a centrifugal compressor goes below a certain throughput at a given head, the performance of the compressor becomes erratic, and the vibrations associated with this erratic behavior can be extremely dangerous. To prevent this, most centrifugal compressors have a protection mechanism that ensures a minimum throughput by opening a recycle valve.

As the drop in throughput can sometimes be very fast, the controller must react quickly when surge is approached and control smoothly when the compressor is operating normally.

The surge controller is an extremely rapid acting controller. The control algorithms used to prevent a compressor from going into surge are extensions of the usual PID control algorithms.

21. Add a valve named as **Kickback Valve** to the flowsheet and define a valve opening percentage of **0%** on the **Rating | Sizing** page. This valve will be sized later once it gets fully solved.



Valve: Kickback Valve

Design Rating Worksheet Dynamics

Rating

Sizing
Nozzles
Options
Flow Limits

Valve Operating Characteristics

☒ Linear
☐ Quick Opening
☐ Equal Percentage
☐ User Table

Valve Vapor Flow Models

Universal Gas Sizing

Size Valve

Sizing Conditions ☒ Current ☐ User Input

Inlet Pressure [kPa]	<empty>
Molecular Weight	<empty>
Valve Opening [%]	0.00
Delta P [kPa]	<empty>
Flow Rate [kg/h]	<empty>

Sizing Methods ☒ Cv ☐ Cg

☐ ANSI/ISA method
☒ Manufacturer specific methods
☐ Simple resistance equation

C1	33.5
Km	0.9000
Cv [USGPM(60F,1psi)]	<empty>
Cg	<empty>

Delete Requires a feed stream Ignored

22. Double-click on **Stage 1** and go to the **Dynamics | Specs** page.

23. Add a Surge Controller by clicking on the **Create Surge Controller** button.

Compressor: Stage 1

Design Rating Worksheet Performance Dynamics

Dynamics

Dynamic Specifications

Duty [kJ/h]	1.757e+007	☐
Adiabatic Efficiency	84	☐
Polytropic Efficiency	84	☐
Pressure Increase [kPa]	1816	☐
Head [m]	2591	☐
Fluid Head [kJ/kg]	25.41	☐
Capacity [ACT_m3/h]	8899	☐
Speed [rpm]	5000	☒
Linker Power Loss [kJ/h]	<empty>	☐

☐ Reciprocating (Positive Displacement)
 ☐ Screw Compressor
 ☒ Use Characteristic Curves
 ☐ Electric Motor

Calculate Head

Create Surge Controller...

24. Set the Output Target Object for the controller as the Kickback Valve | Actuator Desired Position.

Stage 1: Surge Controller

Connections Parameters Monitor User Variables

Name: Stage 1: Surge Controller

Process Variable Source

Object: KO Vap

Variable: Actual Volume Flow

PV → OP

Optional -- Upstream Surge Controller Output

SP

Output Target Object

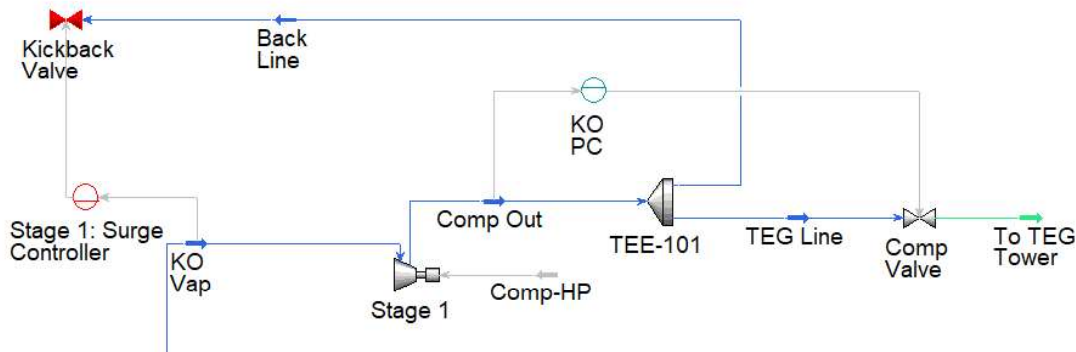
Kickback Valve

Select OP...

Missing constant (A)

Delete Face Plate... Control Valve...

25. Disconnect the compressor discharge stream (**Comp Out**) from the compressor discharge valve (**Comp Valve**). Insert a tee and connect **Comp Out** as the inlet to the tee.
26. Next, add two outlet streams to the tee; **Back Line** and **TEG Line**. Connect **TEG Line** as the feed to the **Comp Valve**, and **Back Line** as the feed to the **Kickback Valve**.



27. Connect the outlet of the **Kickback Valve** to the inlet of the **Knockout Drum** separator using a stream named as **KO Return**.

Note that the surge controller is used exclusively for compressors. When the controller operates in Auto mode, the set point will be calculated by the controller itself and will not be set by the user.

The surge curve of a compressor can be explicitly entered in tabular form as a Speed vs. Flowrate curve on the Rating | Flow Limits page of the compressor property view. For the purpose of configuring the surge controller, the surge data must be fitted against the following quadratic polynomial:

$$Head(m) = A + B \cdot [Q(m^3/s)] + C \cdot [Q(m^3/s)]^2$$

In addition to the surge line, the user must also specify the control line and the backup line, which are described below:

Control Line

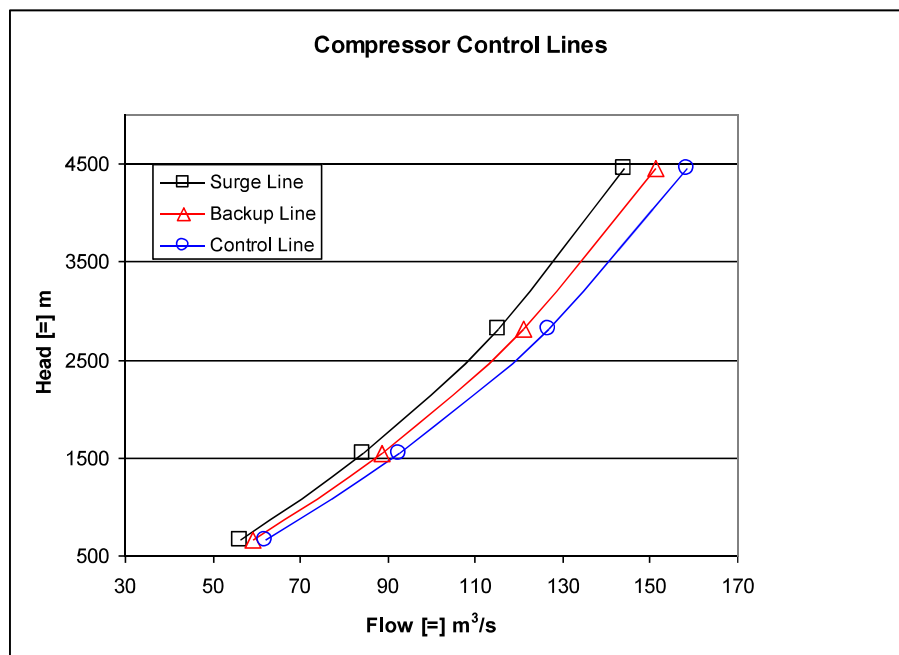
The control line determines the “set point” for the surge controller. This line is set at some fixed percentage above the surge flow (typically 10%) and the controller tries to maintain the compressor flow above this control line. If the flow through the compressor is above the backup line (described below) then the control action is the “normal” PID action. The control line set point can be determined by multiplying the flow from the surge line equation by $(100+x)/100$, where x is the flow % above the surge line:

$$Q(\text{m}^3/\text{s}) = \frac{100 + x}{100} \sqrt{\frac{\text{Head}(\text{m}) - A}{C}}$$

Backup Line

The backup line is located in between the surge and control lines. If the suction flowrate is above the backup line, then a standard PID algorithm is used. However, when the flow goes below the backup line, more aggressive action is taken to prevent surge. Similar to the control line, the backup line set point can be determined by multiplying the flow from the surge line equation by $(100+x)/100$, where x is the flow % above the surge line.

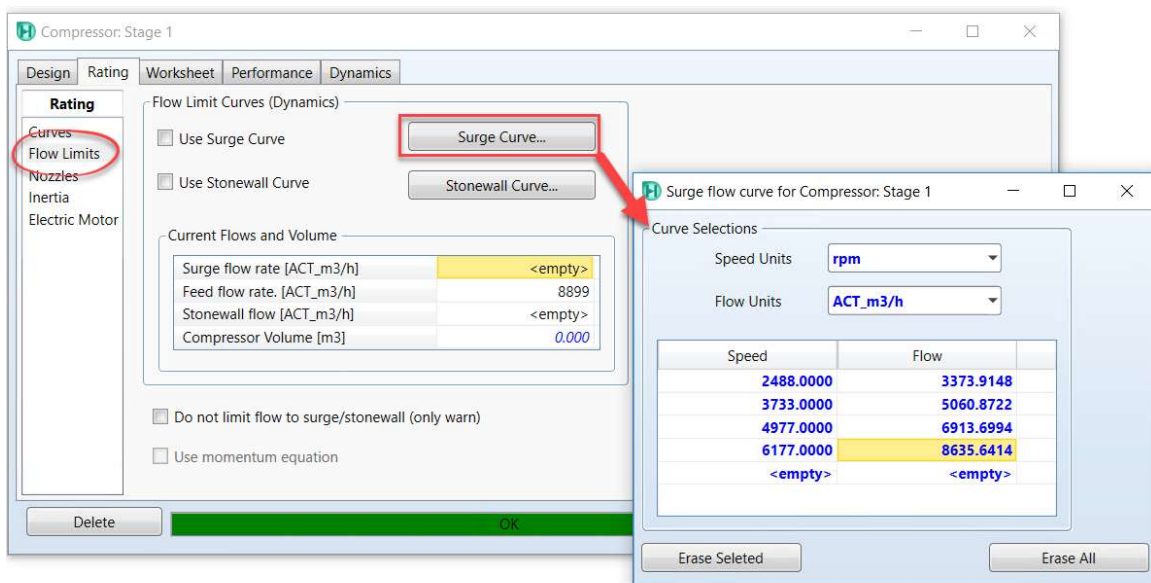
The relative position of the three lines is illustrated in the figure below. For a given head, the flowrate can decrease until it hits the control line, then the surge controller starts to open the surge valve, using the regular PID mechanism. If the flowrate continues to drop and reaches the backup line, the backup control mechanism replaces the regular PID mechanism and more aggressive action is taken (i.e., the quick opening algorithm replaces the PID algorithm and the recycle valve will be opened rapidly at a specified fixed rate).



28. Look at the curves defined for the **Stage 1** compressor by navigating to the **Rating | Curves** page and clicking on the **View Curve** button after selecting one curve in the table.
29. View each curve and fill in the following table using the lowest flow data points from each curve. Be careful with the volumetric flowrate units of measure.

Speed	Min Vol Flow (m3/hr)

30. Go to the **Rating | Flow Limits** page. Click on the **Surge Curve** button. Fill out the table. Be careful with the units of measure. If you input the limit data here, then the surge controller will be able to determine the surge curve polynomial coefficients (A, B and C) by using these data.

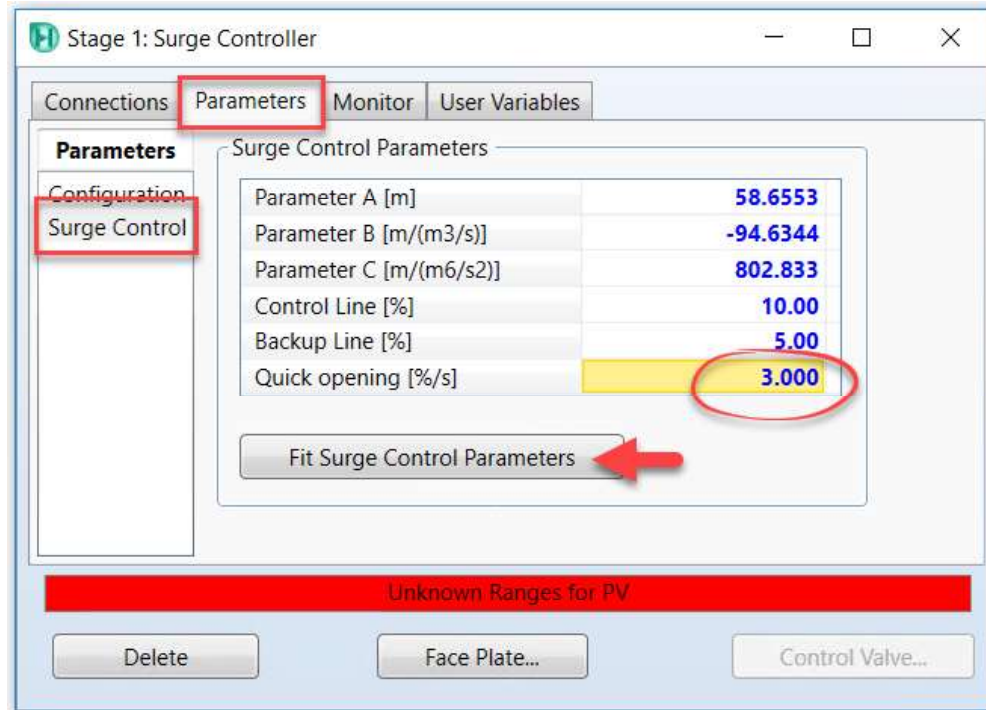


31. Double-click on the Surge Controller on the flowsheet and go to the **Parameters | Surge Control** page.

32. Click on the **Fit Surge Control Parameters** button. It will regress the A, B and C coefficients.



Note: If you don't fill out the Flow Limits form, the Fit Surge Control Parameters button will be inactive. Why? Aspen HYSYS does not assume that the lowest flow points from each curve belong to the surge curve. In reality, they may or may not.

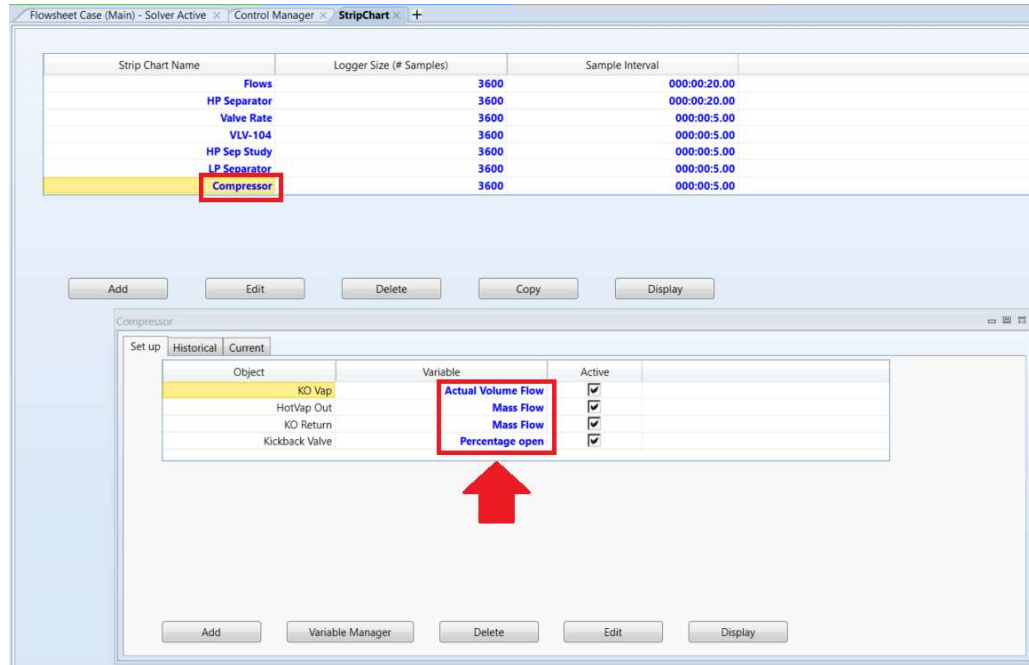


33. The **Control Line** and **Backup Line** are set at 10% and 5% above the surge line, respectively; keep these defaults. The recycle valve will be opened at a rate of 3% per second, as shown in the image above.
34. Go to the **Parameters | Configuration** page and enter the tuning parameters and PV range as shown below:

Controller Settings	
Parameters Configuration	
Action	Reverse
PV Minimum	0 m3/hr (0 bbl/day)
PV Maximum	20,000 m3/hr (3,019,104 bbl/day)
Kc	0.25
Ti	0.1 minutes

35. Set the controller operating mode to **Auto**. Generate a **Face Plate** for the controller if you want to.
36. Size the **Kickback Valve** such that its C_g is equal to 30% of that of the **Comp Valve** C_g value.

37. Build a **strip chart** to monitor the **Actual Volume Flow** for the compressor suction flow (**KO Vap**), the net flow through the system (**HotVap Out**), the kickback flow (**KO Return**) and the **Kickback Valve | Percentage open**.



38. Run the integrator for a short while to ensure calculations are carried out normally and there are no missing inputs.
39. Decrease the flow set points of **FC-Alpha**, **FC-Bravo**, and **FC-Charlie** gradually to starve the compressor of suction flow and force the surge controller into action.
40. Run the integrator again and observe that the valve opens as the control line is hit. If the backup line is hit, one should be able to see the valve opening go from PID control to the Quick-Opening algorithm.

